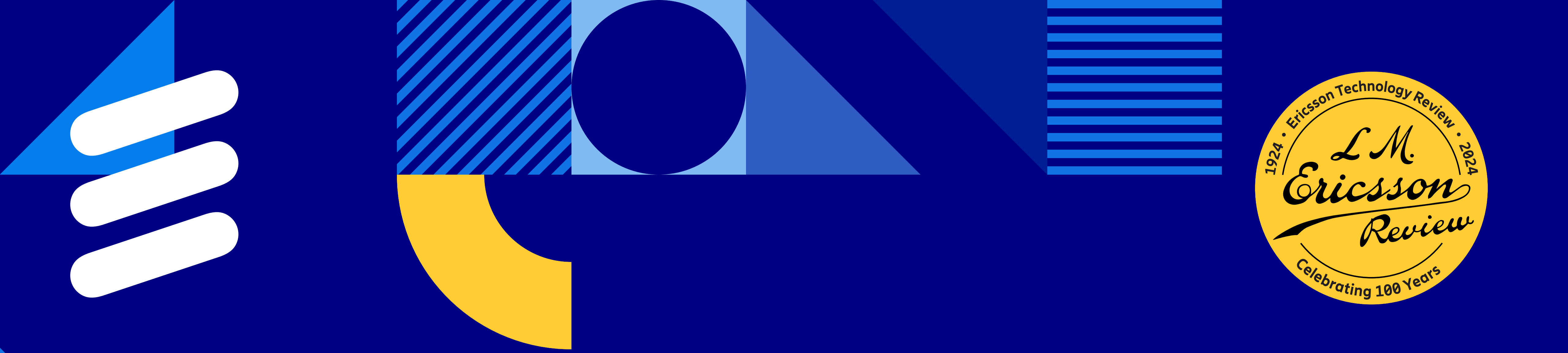
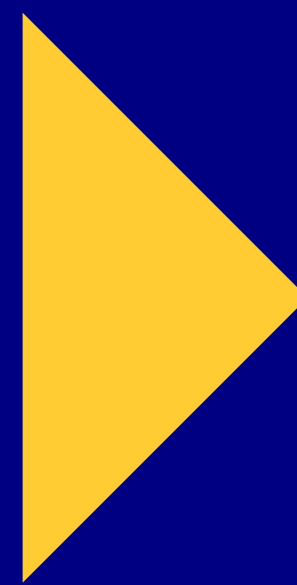




Ericsson Technology Review

Charting the future of innovation



Volume 113 | #02 2024

CTO tech trends 2024

Beyond bit-pipes — new
opportunities with 6G

Ericsson and the future of
telecommunications



Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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CTO TECHNOLOGY TRENDS 2024: Key trends shaping the future of digitalization

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Beyond bit-pipes – new opportunities on the 6G platform

The 6G platform presents new monetization opportunities for communication service providers by making it much easier to provide additional value-adding services that go beyond communication, as well as speeding up the introduction of differentiated/customized communication services.



Charting the path toward the platform economy

This year marks the 10th anniversary of Ericsson’s CTO tech trends article series whose purpose is to anticipate technology evolution, highlight leading use cases, and, in more recent years, flesh out those use cases with business innovation. Today, with the telecom industry on the cusp of a massive transformation toward a platform economy, service innovation is a top priority.

To ensure the cost efficiency, sustainability and trustworthiness of the high-performing networks that will soon be required, Ericsson has prepared a powerful toolbox equipped with a developer platform, network application programming interfaces (APIs), network exposure and network slicing, in order to enable differentiated services.

This year’s tech trends article, on page 26 of this issue of the magazine, explains the key trends driving the evolution of high-performing networks:

1. Using network programmability to enable service differentiation
2. Leveraging telecom AI to boost programmability
3. Transitioning toward a network architecture with horizontal layers to improve and expand capabilities
4. Pursuing innovations over the next decade that enhance performance, sustainability and security.

The seven other articles contained in this issue cover topics ranging from energy efficiency and intent-based service management to cloud-native observability and hybrid core network architecture. We have also included an article that explores the new opportunities available on the 6G platform. The 6G platform will further expand the scope of networks



beyond communication services to support the services needed by applications, which are at the center of the digital transformation. Applications will benefit from the 6G platform by gaining easy access to network capabilities and a wide range of service APIs. Communication service providers (CSPs) will benefit from an expanded role in the ecosystem with new possibilities for monetization higher up in the value chain. Consumers will benefit from improved quality of experience. The article on page 47 presents the 6G platform concept, demonstrates how services are exposed as APIs and concludes with a promising use case example.

It is increasingly clear to everyone in our industry that autonomous networks have great potential to reduce network operation costs and open up lucrative opportunities to address new customer segments and business cases that would benefit from flexible, scalable and customizable service offers. We believe that the introduction of intent into

service management is a critical step on the path toward the creation of autonomous networks. To ensure a cost-efficient transition, we recommend the implementation of intent-based service management for a prioritized subset of services where CSPs expect intent to have the greatest business impact, while continuing to handle other services in the traditional way. Find out more by reading the article on page 19.

We hope you enjoy this issue of our magazine and that you will share our insights with your colleagues and business partners. You can find both PDF and HTML versions of all the articles at: www.ericsson.com/ericsson-technology-review

Erik Ekudden
Senior Vice President,
Chief Technology Officer

100 years of Ericsson Technology Review: Five early insights that became world-changing innovations

Author:
Hans Bergström

The art of timing is everything in innovation. For ideas that arrive far ahead of market maturity, innovation becomes a game of patience and perseverance – as these insights from early Ericsson Technology Review articles demonstrate.

Parallels to the past



The 20th century was one of huge transformation shaped by war, peace and massive economic disruption. Together with motorization, electrification, and digitalization, the evolution of new technologies within telephony, and later mobile telephony, has played an important role in driving key technological advances.

Throughout this period, Ericsson Technology Review (ETR) has been at the heart of this change – spreading new ideas, opening possibilities for new technology paradigms, and documenting an impressive trail of technology leadership on everything from telephone exchanges [1] to extended reality [2].

100 years of innovating ahead of the curve

Timing plays a decisive role in the innovation life cycle. History is full of examples where it has sometimes taken decades before an inflection point occurs. Ericsson founder Lars Magnus Ericsson established this pedigree already in 1896 with the world’s very first wall-mounted telephones – years ahead of the rest of the industry [3].

The ETR archives tell similar stories: in the early 1920s, ETR showcases many early prototypes of concepts we now take for granted, demonstrating an intuitive insightfulness as to the wider direction of technological travel. Some of these

inventions were brought to market by Ericsson; others were leveraged by competitors or jointly developed for the benefit of society.

Below, we pay homage to 100 years of technological intuition inside the pages of ETR and its ability to relentlessly scout technology trends years ahead of time.

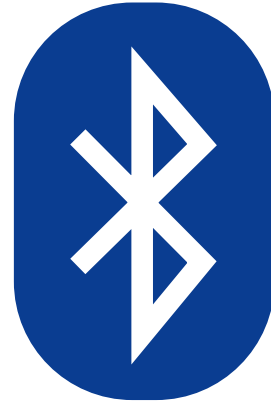
First steps toward Bluetooth in 1924?

Most people today enjoy complete untethered freedom to walk and talk through a hands-free device without the constraints of a cord.

The problem with cords was addressed already in the first issue of the L.M. Ericsson Review [3] with wireless headphones marketed by Ericsson, as shown in **Figure 1**.

The headphones were praised for their lightness (0.4kg) but with battery technology lagging there was still one crucial piece remaining: a lightweight battery. So, with the requirement of a power cord, maybe the headphones were not entirely wireless after all.

When battery technology evolved in the late 1990s, Ericsson finally accomplished its near 80-year Bluetooth quest [4]. On the back of Ericsson’s invention of Bluetooth technology, a community of companies set probably the most widely used standard in wireless sound today.



Terms and abbreviations

3GPP – 3rd Generation Partnership Project | **5GC** – 5G Core | **gNB** - gNodeB | **NG** – Interface between the gNB and the core network | **NMS** – Network Management System | **NTN** – Non-Terrestrial Networks | **RF** – Radio Frequency | **SNO** – Satellite Network Operator | **UE** – User Equipment | **Uu** – Interface between the gNB and the UE |





R 101

THE ERICSSON WIRELESS HEADPHONES.

We are now producing a special wireless model of headphone, combining **lightness** with **strength** and **durability**, with a simple adjustment allowing it to fit comfortably to any head.

The double pole watch receivers have cases of aluminium, and are flexibly held together by leather covered steel bands, all other metal parts being of nickel plated brass.

The headphone is supplied with a soft, flexible cord, a metre and a half long.
Total weight: 0,4 kgs.

We manufacture these Phones to the following resistances:
RF 1320. Resistance, **2000 ohms.** Code word: Repmakvoel.
RF 1321. Resistance, **4000 ohms.** Code word: Repmakvoid.

ALLMÄNNA TELEFONAKTIEBOLAGET
L. M. ERICSSON
STOCKHOLM.

Figure 1: One of the first prototypes of cordless headphones from L.M. Ericsson Review 1924. Bluetooth would arrive nearly 80 years later with the evolution of battery technology.

Parallels to the past

Speaking in the rain...or even under the sea

During the late 1890s, Ericsson invented a watertight microphone [5], as shown in **Figure 2**. Today, we expect our phones to be (at least somewhat) waterproof, which has not always been the case. With the arrival of the R310 “shark fin” phone in 2000 (Figure 3), Ericsson led the way more than 100 years after the invention of the watertight microphone.

In the mid-1930s, Ericsson even tested the waters in diving equipment with a range of field tests to demonstrate subaquatic communication, as shown in Figure 4.

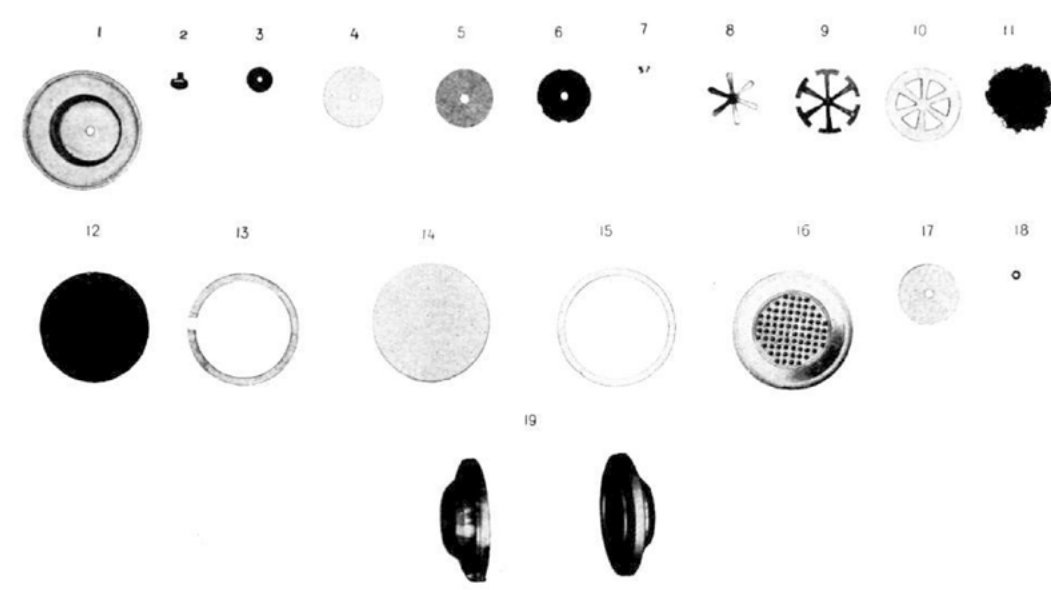


Figure 2: An early design of a watertight microphone (left) invented already in the 1890s.



Figure 3: The Ericsson R310 ‘shark fin’ mobile phone that appeared over 100 years after Ericsson’s early design of a watertight microphone.

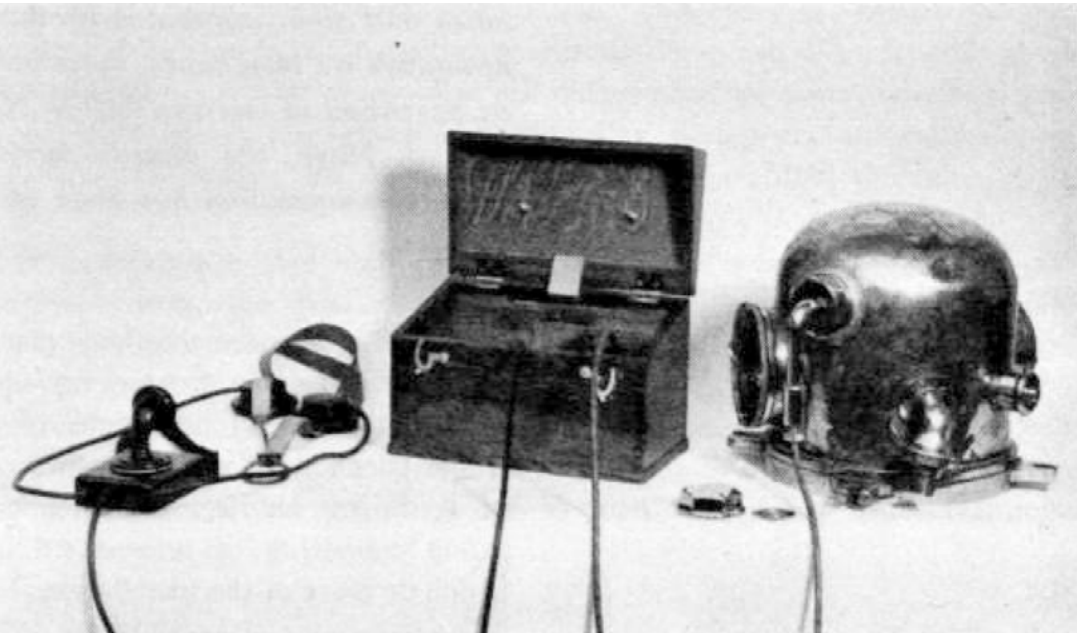


Figure 4: Communication diving equipment from the Ericsson Review in 1935 [6].



Figure 5: The first transatlantic video telephone conversation exchanged between Washington and Sweden at the beginning of December 1971 [7]. The photograph was taken at the COMSAT building in Washington.

Markets reject video communication in the 1960s

In the 1960s, Ericsson took its first significant steps into developing videophone capabilities, culminating in the first transatlantic video call in 1971 as shown in **Figure 5**. Despite the development of these early prototypes with enterprise markets in mind, all efforts would prove in vain with the markets showing a reluctance to accept such a breakthrough.

It took 60 years and a global pandemic to largely reverse that reluctance. Today’s more integrated devices make for a much smoother adoption of video communication. In fact, having dedicated terminals may have hindered early market development. Now more than half a century later, everyone’s laptop and smartphone has these capabilities.



Figure 6: Video communication has become a common tool both socially and at work in the post-pandemic era.

Parallels to the past

Shopping on-the-wire in the 1920s: an early contender for e-commerce

Online retail has reshaped the world of commerce, not to mention adjoining sectors across the value chain.

However, already in 1926 [8] we begin to see its early origins in ETR – not so much online retail, more “on-the-wire” retail. As mass telephony began to gain a foothold, Ericsson was quick to point out the benefits of telephone networks as a tool to make retail easier and more convenient, as you can see in **Figure 7**.

Fast forward to 2000 and ETR [9] was already laying the groundwork for online shopping using mobile devices, among other predicted use cases, almost ten years before the evolution of the smartphone as we know it. **Figure 8** illustrates some of those early predicted use cases.

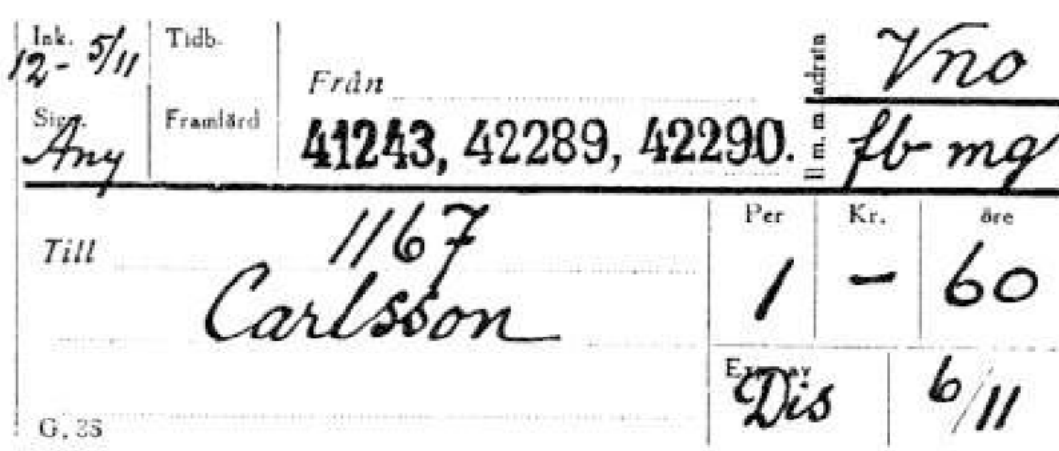


Figure 7: In 1926, Ericsson was quick to capture the benefits of “on-the-wire” retail. In this example, market traders would use the telephone to place and receive fish orders remotely. Cod anyone?

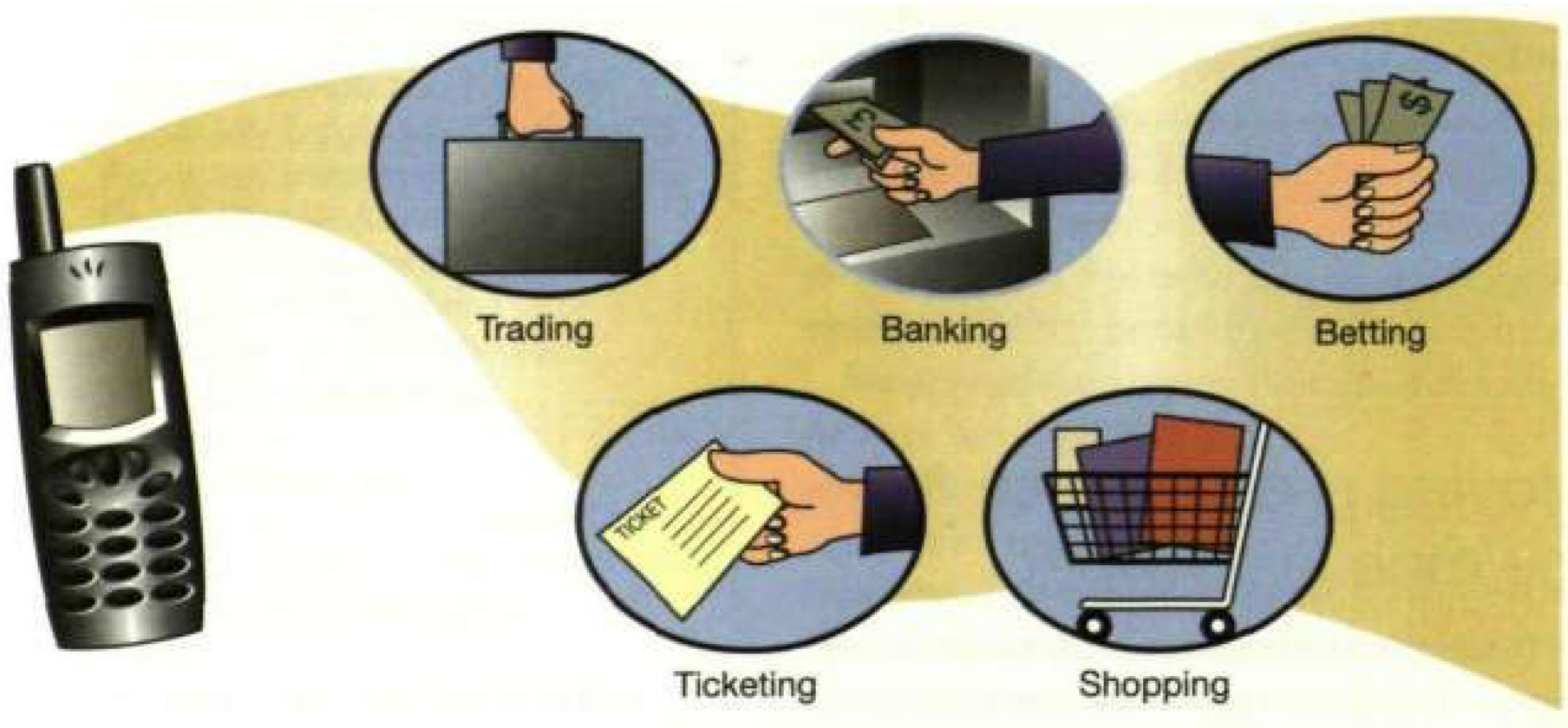


Figure 8: By 2000, almost ten years ahead of e-commerce using a smartphone, Ericsson was already successfully predicting future use cases including mobile shopping and mobile banking.

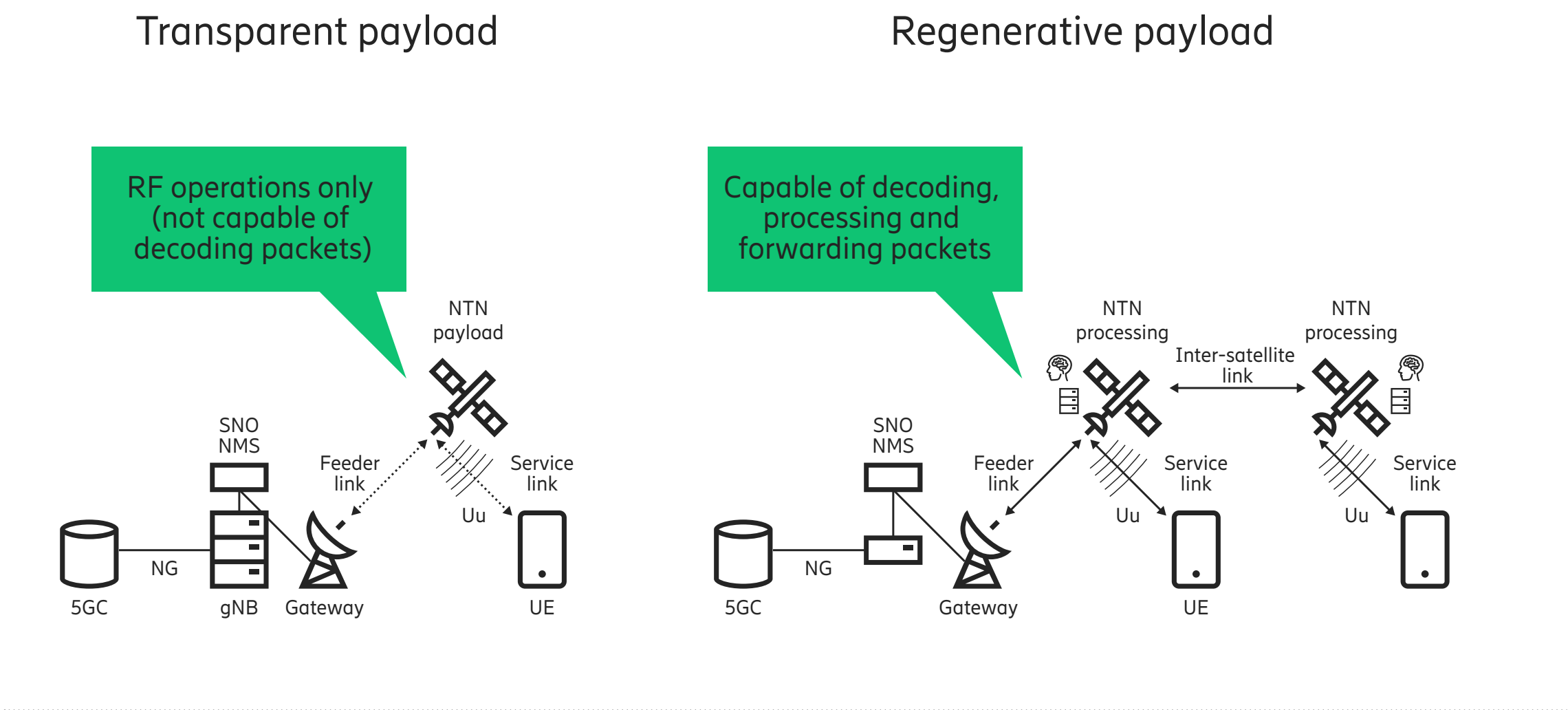


Figure 9: An overview of existing satellite systems from ETR, 2023.

Satellite communication: always in fashion

Non-terrestrial networks (NTN) became part of the 3rd Generation Partnership Project (3GPP) standard in Release 17, establishing a strong foundation for direct communication between satellites, smartphones and other types of mass-market user equipment [10].

However, this isn't the first time our industry has become enthralled by the prospect of sending satellites into space. A look through the ETR archives tells you that the interest has long been there.

Already in 1964, ETR [11] describes a transistorized carrier terminal for the transmission of 12 telephony channels on radio links along with TV transmissions from the US to Sweden, all in collaboration with NASA.

In the late 1990s, there were some attempts to introduce satellite communication to mobile telephony, with Ericsson producing a number of satellite phones. This was one of many stepping stones that evolved into NTN, part of the 3GPP's Release 17 since 2022.

ETR brought such insights to a broader public, with an article dedicated to satellite communication with 3GPP technology [12], as **Figure 9** illustrates. The article explains how communication service providers could cooperate with satellite operators to jointly create a global NTN ecosystem by enabling connectivity between terrestrial systems and satellite systems on the same mobile platform.

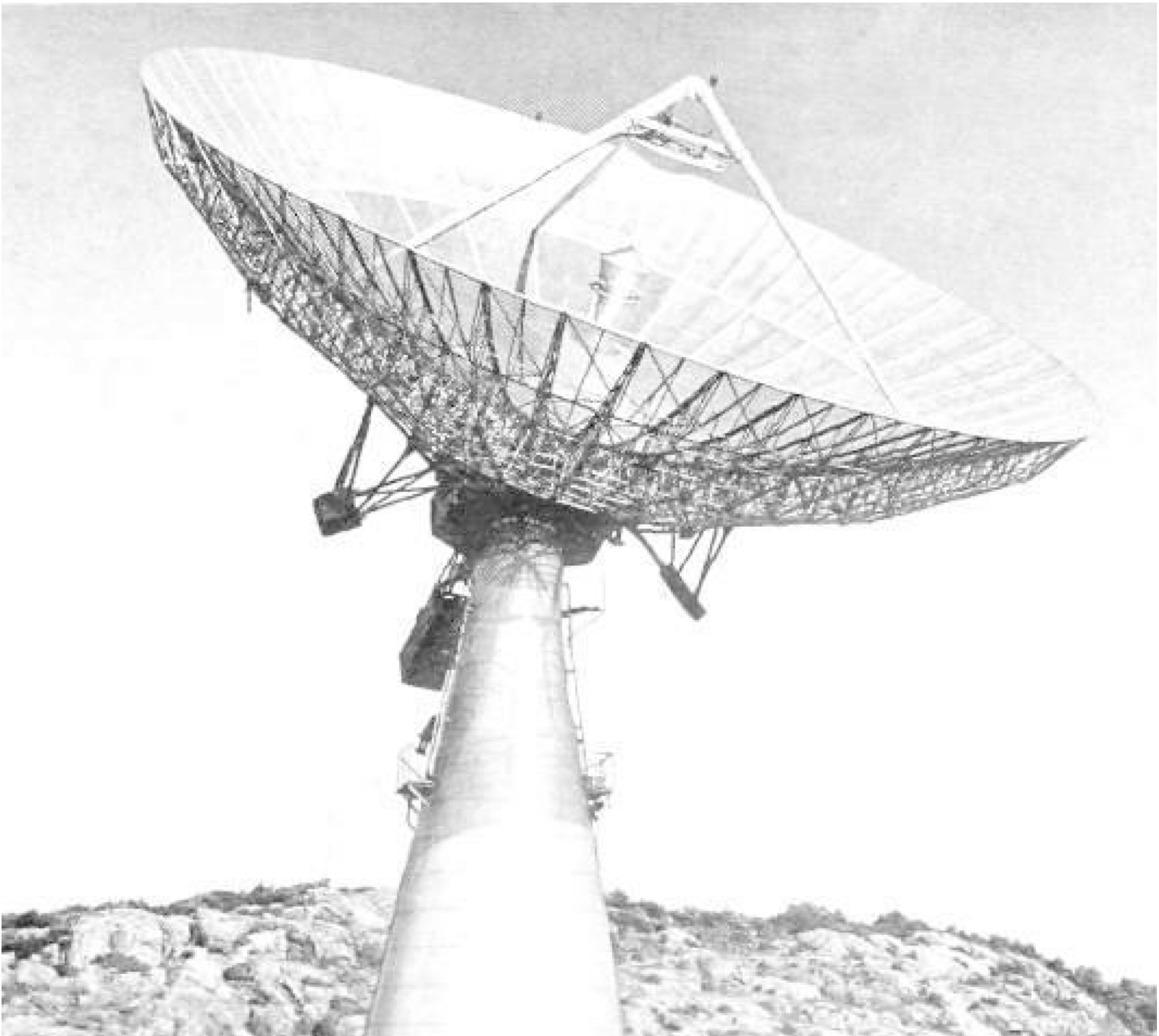


Figure 10: A satellite dish on the island of Råön, north of Gothenburg, from Ericsson Review, 1964.



Figure 11: By harnessing Hollywood, Ericsson brought its innovations to a broader audience. The iconic Ericofon featured in several movies in the 1950s and 1960s.

Harnessing Hollywood to demonstrate leadership

Marketing also plays a part in the innovation life cycle, ensuring emerging solutions align with market maturity. Throughout the years, Ericsson has even adopted an innovative approach to this, turning to Hollywood to demonstrate its technology leadership.

Ericsson’s iconic Ericofon (or Cobra phone) featured in several movies [13], as shown in **Figure 11**. While in the 1997 James Bond classic *Tomorrow Never Dies*, 007 could be seen driving his car remotely with his Ericsson JB988 cell phone (**Figure 12**).

Celebrating 100 years of thinking differently

Looking back at 100 years of insights, all successful innovation has a lot to do with technology; the rest is about timing and business development. With innovation, it is easy to be too early, especially in a large organization like Ericsson which is years ahead of the curve.

For more than 145 years, Ericsson has continuously sought to reinvent itself, exploring many broad, diverse and sometimes even unexpected business areas. In the 1890s, for example, most people would not be able to imagine a waterproof mobile phone, nor could they predict the success of Bluetooth in the 1920s.



Figure 12: The Ericsson JB988 mobile phone played a starring role in the 1997 James Bond classic *Tomorrow Never Dies*.

The tracks of visionary thinking found in the ETR have built a culture of pushing boundaries to imagine the impossible. In the past 100 years, this impossible thinking has laid the foundation for the development of paradigmatic technologies such as mobile broadband, Bluetooth, video calling, immersive communication, non-terrestrial networks, and more.

The authors



Hans Bergström has well over 30 years’ experience within communication services and has extensive global experience in different capacities from customer engagements to architectural and systems design. His background spans several segments including business communication, networks, data centers and wireless communication all through 3G, 4G, 5G and 6G. In his current role as director of Architecture Evolution, he is researching the emerging trend of high performance networks.

Acknowledgements

We would like to acknowledge the contributions made by generations of skilled, innovative and insightful Ericsson colleagues within all domains required to facilitate this monumental paradigm shift.

Parallels to the past



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- Ericsson Technology Review archive ↗
- Communication – the rise of telephony – Ericsson ↗

Ericsson and the future of telecommunications

Authors:
Kristina Gold, Pernilla Jonsson

In 2024, Ericsson Technology Review is celebrating 100 years of showcasing technology breakthroughs and innovations driven by continuous advances in engineering sciences and information and communications technology. What does this wealth of knowledge and experience tell us about what to expect in the next 100 years?

Ericsson and the future of telecommunications



“The best way to predict the future is to create it.” This powerful sentiment, often attributed to Abraham Lincoln, reflects the heart of the Ericsson culture: our mission is to innovate and shape the future.

The astounding pace of innovation that we have become accustomed to today would have been unimaginable in 1924. It took decades before 19th- and 20th-century technology developments such as electricity, cars, telephony and computers had a real business and economic impact, and at least a decade longer before they had any significant social and cultural impacts. In the 21st century, by contrast, the step from technology development to business and social impact has tended to happen within a decade. The introduction and adoption of smartphones is the best example of this. While it took 100 years to connect 1 million places using fixed telephones, it took less than 10 years to connect 1 billion smartphones using mobile communication networks.

Social media platforms and virtual global marketplaces such as app stores are playing a major role in speeding up the dissemination of technology innovation. For example, it is estimated that the generative artificial intelligence (AI) tool ChatGPT took only two months to reach 100 million users, setting the record for the fastest user-base growth in history [1]. Looking ahead, analysts predict an even faster pace for new technologies and business models to reach mass markets.

Ericsson’s innovation engine – built to last

For more than a century, Ericsson has worked tirelessly to create a culture of innovation through technology leadership. Reaching and maintaining a position as an influential actor in the ecosystem requires that we constantly challenge the status quo and push the technology frontier in both the short and long term. Innovation is critical at all steps of the research and development (R&D) process. Ericsson invests sizeable resources in technology foresight – but also in predicting the likely development of economic, social and lifestyle trends, as well as changes to the geopolitical landscape, social media and sustainability goals.

At Ericsson, we are committed to building and sustaining a robust innovation engine that will last for centuries to come. The circle in **Figure 1** visualizes our approach.

A diverse global R&D footprint is an essential component of our strategy, which is why Ericsson’s tens of thousands of R&D employees are strategically based at major technology hubs all around the globe. With this approach, we have access to the best talent in the world, as well as a wealth of local knowledge, new insights and fresh ideas. Our global R&D footprint also provides ample opportunity for research collaborations with leading global centers of excellence in both academia and industry.

Our approach to inventing for the future

Ericsson has always focused on finding new ways to transcend physical space and connect people, which has inevitably led to disruption. Our core business has shifted several times – from repairing phones to developing fixed

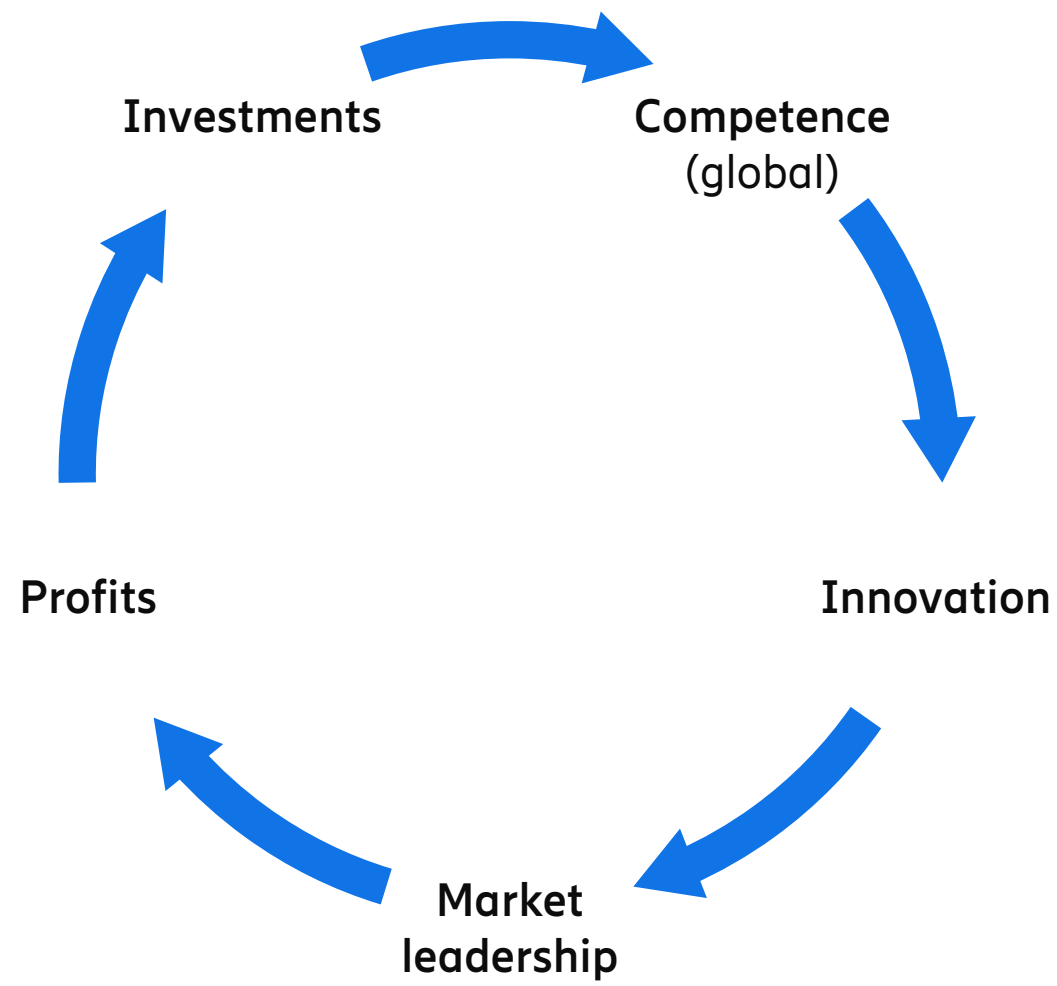


Figure 1: Key components of Ericsson’s innovation engine

telephony, for example, to developing fixed telephony systems and switchboards, before progressing to mobile networks, mobile phones and mobile broadband. The app economy ultimately emerged from the possibilities of 4G in combination with the smartphone, which eventually led to the smartphone becoming a kind of remote control for modern life, providing users with a vast array of services available at the tap or swipe of a finger, wherever you are.

As we work to define the next generation of mobile networks, we have our sights set on a disruptive vision of programmable intelligent networks that offer tailored network services to match the needs of the world. The sweet spot is where technology meets market demand with viable

business models – that’s where the magic happens. To secure a profitable future in the century to come, we know that we must be willing to transform ourselves again and again, allowing the seeds of the next core business to disrupt the core business of today.

But how do you merge business with technology and make the most of the opportunity? At Ericsson, we believe that it is in the interplay between the adoption of evolved technology capabilities and new combinatorial effects that potential business disruptions can be identified. These are typical system-level innovations that can change established business models and potentially even the roles in the industry ecosystem. All of this provides us with important

information about which ICT applications or types of network traffic we can expect in future networks.

As we innovate for 6G, sustainability and digitalization serve as critical pillars. Strong industry momentum in areas such as automation, electrification and immersive experiences will also be important drivers of the requirements of future networks. Concurrently, advancements in AI, cloud, computing and connectivity will be instrumental in unlocking future network capabilities.

Evolution toward the next generation of the internet, in which digital content will interact with the physical world, will profoundly change how human communication needs can be addressed, and is therefore likely to lead to business innovation with the potential to disrupt the ICT ecosystem. On top of providing immersive communication capabilities, it is expected that the next generation of mobile systems will also incorporate Integrated Sensing and Communication (ISAC), which allows reflections of transmitted radio signals to be received in the network and processed to yield spatial knowledge of the physical surroundings [2].

Likely innovation areas in the next few decades

Our century-long experience in the telecommunications space has taught us that the technologies that are most likely to have a big impact decades from now are probably already a decade or two old in R&D terms. One emerging example of this may be quantum computing, which has been a subject of academic research for many years.

Looking ahead toward the second half of this century, we expect to see convergence across a multitude of technology disciplines including electronics, photonics, nanotechnology,

biotechnology, biomedicine, cognition and communication technologies. Rather than converging at component and system level as technologies do today, however, future convergence will happen at the molecular and atomic level, generating an enormous innovation potential to improve society and human lives. Plants, for example, could be integrated with processing capabilities, sensors and actuators to optimize capabilities such as photosynthesis. Quantum communication capabilities could be used to enable plants to communicate their needs to humans or AI bots.

Innovation for the next century – reaching beyond the possible

World-leading innovators must visualize beyond the predictable. 100 years ago, it would have seemed impossible to visit your bank using a device so small it can fit in your pocket. And yet, much of what was physical matter yesterday has become digital today. In the next few decades, we expect that it will be possible to digitalize all five of the human senses, moving beyond sight and sound to include touch, taste and smell.

At some later point in the century ahead, humankind will almost certainly figure out how to break through an even more farfetched-sounding frontier. Wouldn’t it be thrilling if, for example, we could take the first steps toward teleportation by enabling communication to transfer matter from one point to another? Today, with no physical theories for how to transcend time and physical space, it seems impossible. But at Ericsson, more than 100 years of groundbreaking innovations have taught us never to let the word impossible stand in our way.

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Enhancing infrastructure to boost energy efficiency in 5G and 6G core networks

Authors:
Leif Johansson, Per Holmberg, Robert Skog

Communication service providers are keen to reduce energy consumption, enhance performance and reduce the need for large system configurations and power supplies in mobile core networks. To introduce the necessary application knowledge, we have developed a shim layer that can intercept and configure the relevant energy-saving mechanisms in the cloud environment.

Energy-efficient mobile core networks



Energy-saving mechanisms can only work when they are applied correctly. In the cloud environment, application knowledge is essential to achieving energy savings without risking any application key performance indicators (KPIs) or modifying the Kubernetes software.

While Kubernetes-based, cloud-native deployments of mobile core networks deliver a variety of important benefits, they also limit power-management capabilities and hamper energy efficiency due to a lack of knowledge about the execution characteristics and requirements of individual applications. For example, an application running on one node may have lower KPI requirements in terms of latency and performance, while an application on another node is crucial in meeting the characteristic deadlines. If all computers and resources are deployed equally with the same power-management logic and without considering the applications’ needs, the result is likely to be unnecessarily high energy consumption for low-performance tasks and reduced performance when energy is wasted on low-performance tasks.

Overview of energy-saving mechanisms in cloud infrastructure

Figure 1 shows a cloud-native deployment with a guaranteed container (in purple) that has exclusive central processing unit (CPU) cores (Core 0 and Core 1) next to burstable and best-effort containers (in light green)

that share cores (Core 2 and Core 3). Each CPU core can be individually controlled through frequency scaling (P-state), halt states (C-state) and micro-sleeps. The energy consumption of the CPU fabric is controlled through uncore frequency, which affects the performance of all CPU cores as it is a shared resource.

To save energy without compromising application KPIs such as latency, the four available energy-saving mechanisms – core frequency scaling, uncore frequency scaling, micro-sleep and C-states – must be applied differently depending on the needs of each application.

Core frequency scaling

New CPU generations provide the ability to control the frequency on each core independently. Most mobile core applications are event-driven and require fast reactions to external network events to meet service latency requirements and service fulfillment. Measurements from mobile core networks show service requests arriving in bursts to the service handling node. A traffic-idle period can be suddenly followed by a burst of traffic events that require maximum computer performance to meet the service latency requirements.

The host operating system- (OS-)governed core frequency scaling has shown itself to be too slow to react to traffic bursts, which can lead to reduced system performance when enabled. Therefore, it is advisable to disable the host OS-governed core

Terms and abbreviations

API – Application Programming Interface | **CPU** – Central Processing Unit | **KPI** – Key Performance Indicator | **OS** – Operating System | **PCI** – Peripheral Component Interconnect | **QoS** - Quality of Service | **UPF** – User Plane Function



– from 104W to 177W in our example – due to best-effort services. Therefore, it is crucial to optimize the implementation of host-OS-governed power management to ensure appropriate resource usage and prevent unnecessary energy consumption.

Uncore frequency scaling

The performance of the uncore, which includes the internal interconnect and parts of the processor shared by all cores, is crucial for memory and communication-bound applications. The interconnect performance is controlled by adjusting the uncore frequency. And by maintaining a low frequency, matching the actual needs, it saves CPU power.

However, if application characteristics and requirements are not taken into consideration, the default behavior is typically to scale up the uncore frequency just in case, even when a single core scales to high frequency. This often happens unnecessarily for CPU-bound programs that do not generate significant load on the CPU fabric, as well as background activities that are not critical and can use a lower and power-efficient uncore frequency. Even best-effort services can cause false triggering of high uncore frequency and unnecessarily high energy consumption.

Micro-sleep

Today, if an application needs to react quickly to traffic events, it continuously polls event queues in a busy loop, resulting in constant CPU execution at maximum performance and high energy consumption. To avoid such busy looping, micro-sleeps can be used to save energy between traffic events.

Micro-sleep is a technology that enables processor cores to cease execution until something new arrives in an event queue. The time required for waking up from a micro-sleep

state is only a few hundred nanoseconds, allowing the application to save energy while still reacting quickly to new events. This can be particularly useful for packet processing nodes like the user plane function (UPF), which can utilize micro-sleep while waiting for new packets.

Unfortunately, the micro-sleep state cannot be detected by the host OS today. The host OS treats a CPU core that uses the micro-sleep state to save energy as 100 percent busy and never triggers core/uncore frequency scaling.

C-states

Processors provide lighter and deeper sleep modes (C-states) for cores and clusters of cores with shared cache, as well as for the full processor chip where a lighter sleep state, like C1, allows for a faster wake-up recovery but still has higher overhead than micro-sleep. Part of this comes from the software interface where sleep states are managed by the OS, giving context switch overhead and deeper sleep states that provide better power savings that also have further overhead in saving and restoring processor and cache states.

Proof of concept: Introducing application knowledge

In our proof of concept, we configured the Kubernetes CPU manager to provide guaranteed quality of service (QoS) class containers with an exclusive set of CPU cores, as shown in Figure 1. This approach ensures that containers or services with guaranteed performance can react quickly to network events, as the cores are not being used by anyone else. To reduce the complexity and achieve better hardware utilization, we chose a cloud-native deployment model using Kubernetes over bare metal infrastructure [1], which is the de facto standard.

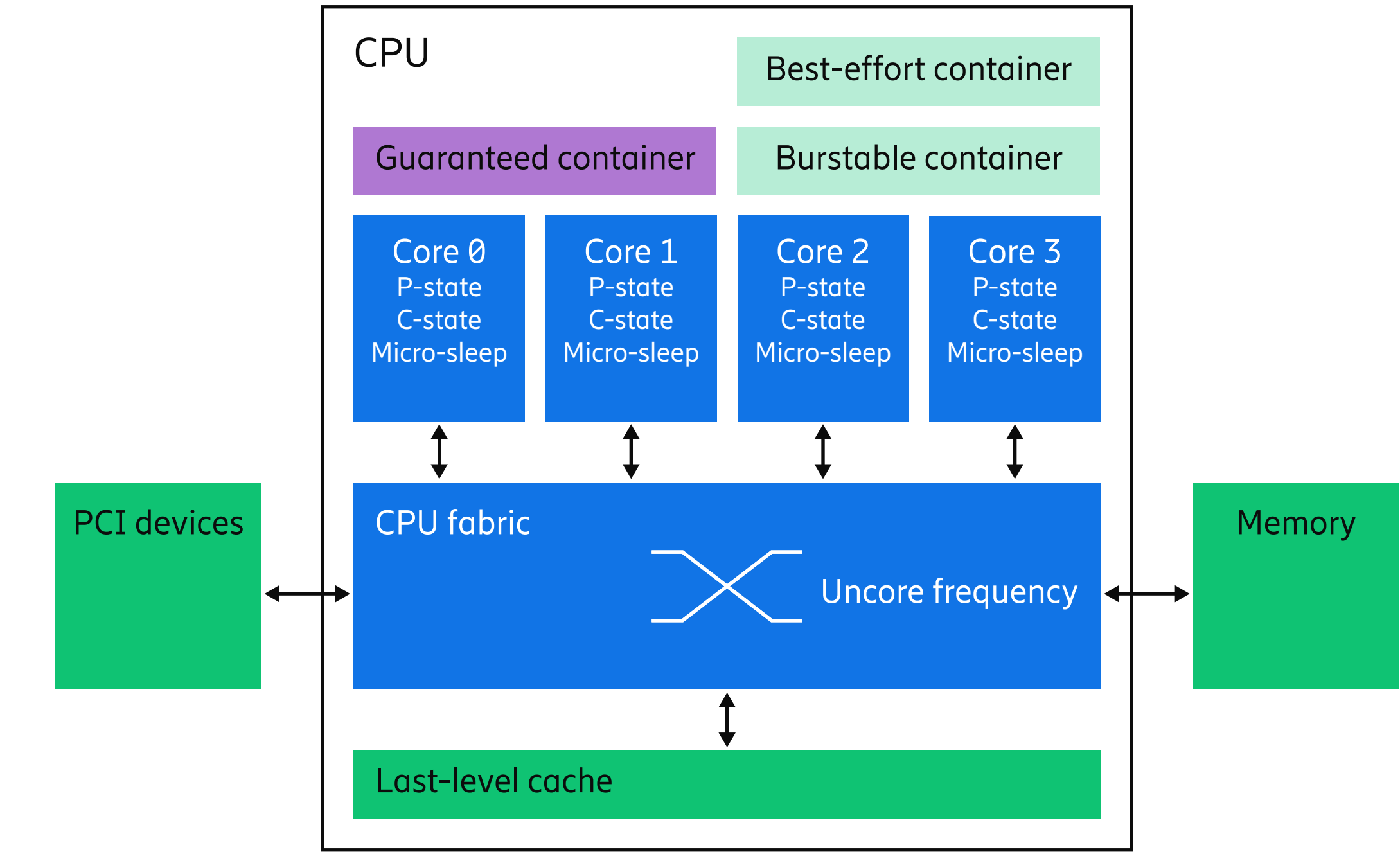


Figure 1: Cloud-native deployment example

frequency scaling when running mobile core applications. Mobile core cloud-native applications consist of several microservices, where each service is described in a chart, such as a Helm chart. Not all services require maximum undisturbed computer performance, as many are best-effort services that can make use of the host OS-governed core frequency scaling to save energy when the service is not in use. However, the host-OS-managed power core frequency treats all microservices equally and cannot distinguish between services that handle traffic and those that do not

require high performance. As a consequence, best-effort services can cause false triggering of high core frequency and unnecessary energy consumption. To reduce energy consumption, it is important to configure the host OS not to scale up the performance for best-effort services, as this unnecessary high frequency results in higher energy consumption.

The results of the proof of concept presented later in this article indicate a notable increase in energy consumption

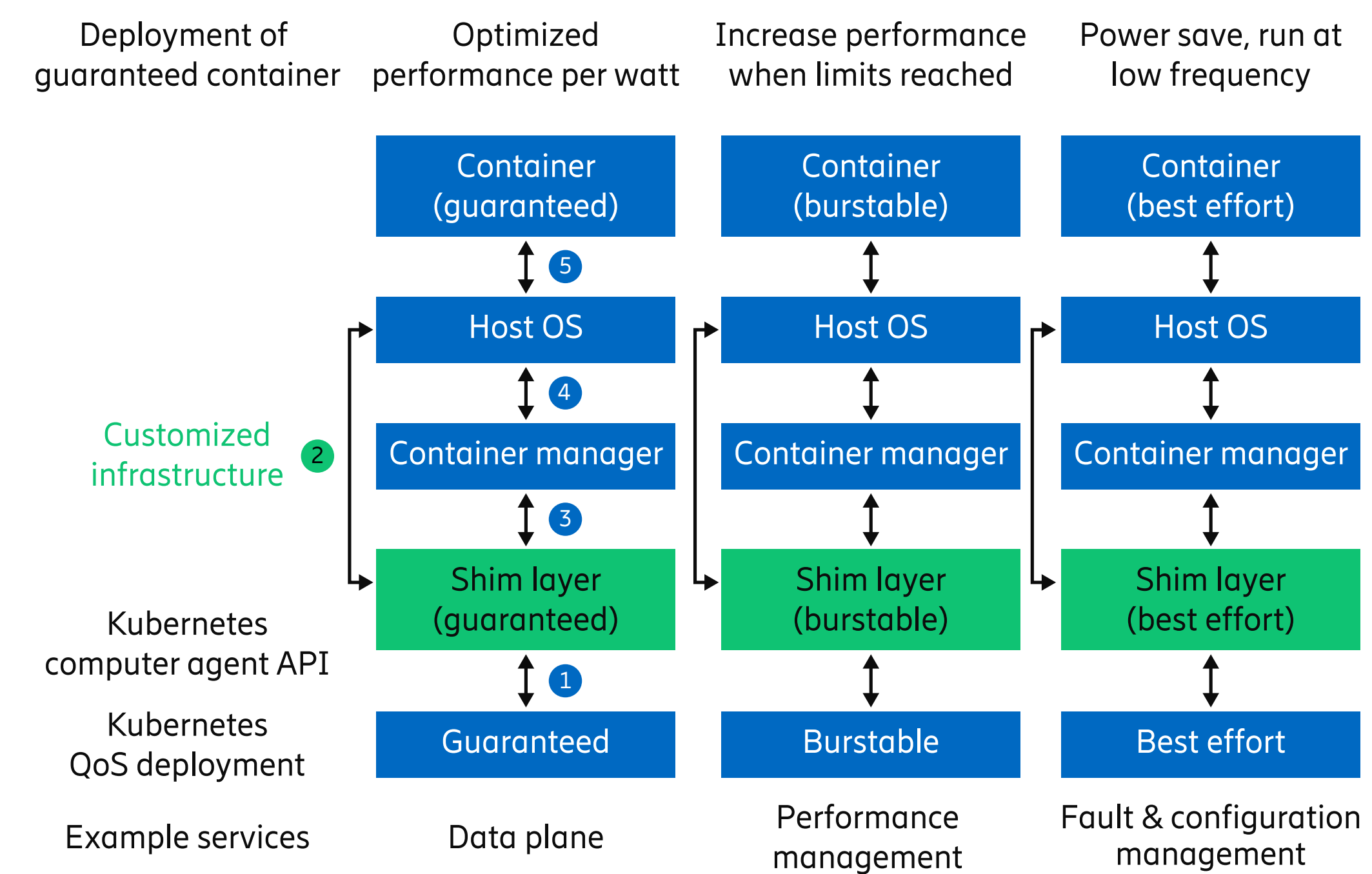


Figure 2: How the shim layer intercepts the container manager socket API

The Kubernetes application package manager we opted for is Helm, where the application deployment is described in Helm charts. The Helm charts contain various infrastructure requirements for each microservice, such as memory usage and CPU resources. Based on real-time characteristics, three different QoS classes can be specified in a Helm chart. Furthermore, a Helm chart can optionally include application hints to optimize power usage. In our research, we have added new application hints to configure the hardware infrastructure.

To achieve the goal of greater energy efficiency when deploying telecom-grade services on cloud-native infrastructure [1], we used a standard Kubernetes container deployer plus a new shim layer, as illustrated in **Figure 2**. The shim layer does not affect Kubernetes and is used to customize the infrastructure power-management functions before and after the container deployment, meeting KPI requirements for critical applications while optimizing energy efficiency.

Our method is based on Kubernetes QoS classes and additional information provided in application Helm charts to deliver application knowledge and application hints. Guaranteed QoS class is assumed in our experiment for services used in traffic execution that need deterministic performance to fulfill application KPIs. The burstable/best-effort QoS is used when the performance requirements and fast response times are more relaxed.

Our focus is therefore to reduce the energy consumption for services with more relaxed requirements (burstable and best-effort containers), and use the saved energy to boost the peak performance of guaranteed containers at traffic peaks, while simultaneously reducing energy consumption when the traffic is low. The shim layer then uses this information to configure the host OS power management, resulting in improved performance and energy efficiency of computer resources, while maintaining the application KPIs.

Setting up the shim layer

In the initial step, the shim layer is configured to customize the infrastructure based on the service type (guaranteed, burstable, best-effort and so on). The socket application programming interface (API), which Kubernetes uses to trigger the start of a new container, is redirected to the shim layer by changing the transport protocol port number used by the container manager to the port number used by the shim layer, marked as step 1 in Figure 2. The port number to start a container is part of the Kubernetes configuration files and can be changed by restarting the Kubernetes services.

Customizing the infrastructure

Before starting the new container, the shim layer starts to customize the infrastructure based on the Kubernetes QoS class and application hints provided in the application Helm

charts, marked as step 2 in Figure 2. As an example, for a guaranteed container, we customized the infrastructure to set the frequency of the guaranteed non-shared CPU cores and uncore frequency as high as possible using the host OS API, to boost the performance for the guaranteed service.

We configured the shared CPU cores used for best-effort/ burstable services to be managed by the host OS and execute at the lowest possible core frequency and at the lowest possible CPU halt state (C-state) when the cores are idle. The shared CPU cores are also configured to never enter opportunistic, not guaranteed frequencies (turbo frequencies) when busy to further save energy. This enables the system to save as much energy as possible for services that are not guaranteed and avoids unnecessary scale-up of the frequency for non-critical services.

For best-effort/burstable configured containers, we forwarded the request to the container manager without any change to the infrastructure, which was already in the lowest power-manager state.

Kubernetes scales burstable containers based on resource utilization during runtime. The resource scaling request is intercepted by the shim layer, and it triggers new infrastructure settings based on the resource demands. For example, when resource demands increase, it will scale up the CPU core frequency.

Application hints

The optionally provided application hints help the shim layer prioritize critical application resources. In our work, we have introduced two application hints: prioritized core frequency and prioritized uncore frequency.

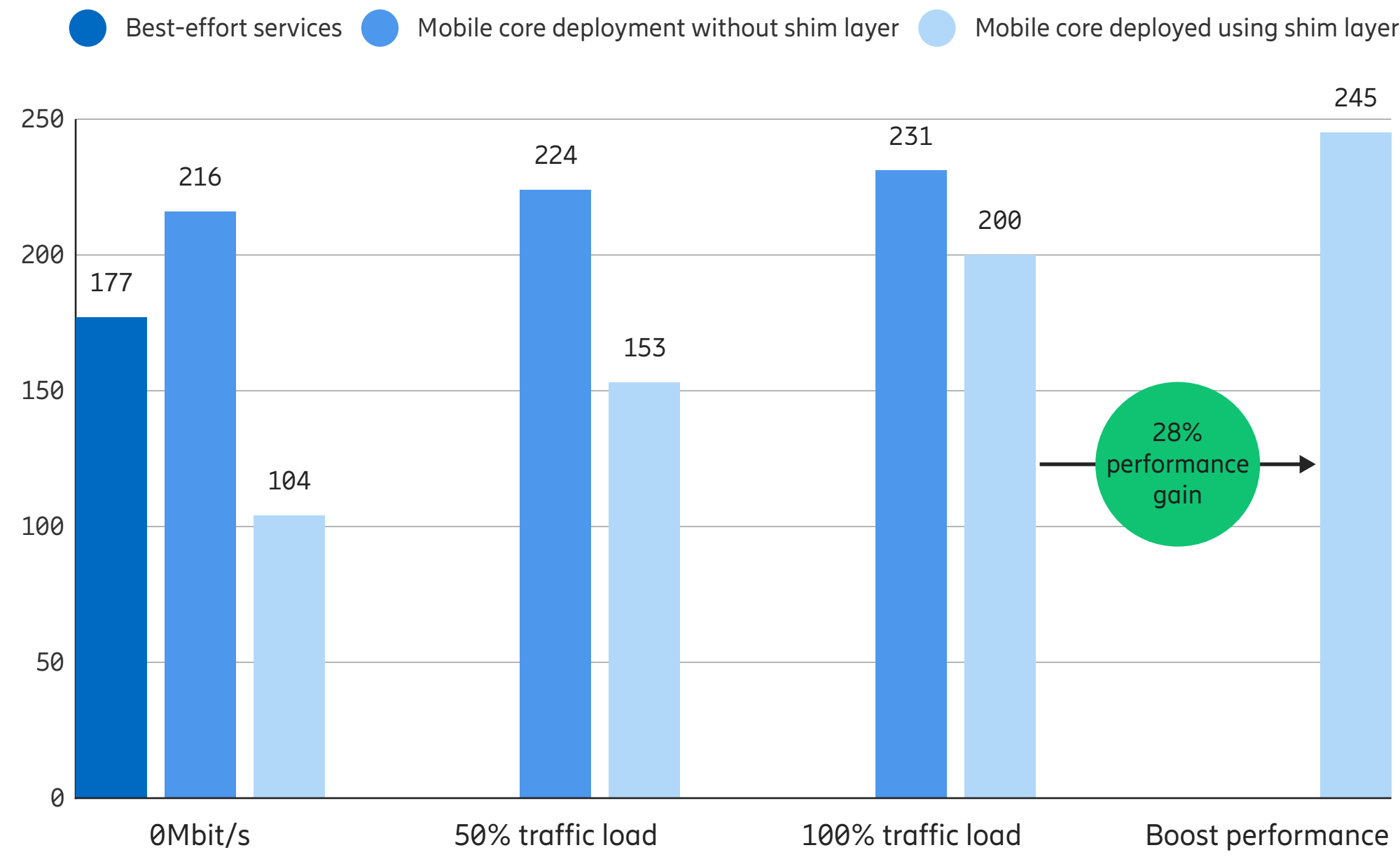


Figure 3: Mobile core CPU power consumption

The core frequency hint is used when the application performance scales with the CPU core clock frequency and is less dependent on the uncore frequency. To achieve maximum performance or improve performance per watt, the shim layer reduces the uncore frequency and scales up the CPU core frequency when the core frequency hint is applied.

The uncore frequency hint is used when the application scales with uncore frequency – for example, applications streaming data from the memory system. For uncore frequency application hints, the shim layer reduces the core frequency and scales up the uncore frequency.

Saving energy at low traffic

To save energy and react quickly to traffic events, micro-sleep is used as soon as the CPU core is idle when running user-plane containers with guaranteed class of service [2]. For services where network events are provided by the host OS and configured with guaranteed QoS class, we configure the host OS to use the fast CPU halt state (C-State 1E) as soon as the guaranteed non-shared core is idle. To further save energy, the core frequency is scaled down to its minimum when the core is idle and scaled up to its maximum core frequency when active (rush-to-idle policy). This enables the application to trigger maximum energy savings when idle and deterministic high performance when active.

The method can be applied without affecting any application KPIs.

Uncore frequency scaling

To avoid high CPU fabric energy consumption, only services with guaranteed QoS class trigger the uncore frequency. In our tests, the guaranteed non-shared CPU core with the highest utilization determines the uncore frequency. Note that the CPU fabric is a shared resource, and the uncore frequency impacts interconnect performance on all CPU cores. The method ensures that non-critical services do not trigger unnecessarily high CPU fabric energy consumption, while critical CPU cores receive the needed performance to fulfill application KPIs.

Forwarding the container request

After the infrastructure customization, the container request is forwarded to the container manager, indicated as step 3 in Figure 2. The guaranteed service is started on the cores assigned to the container (shown in steps 4 and 5). The system is now performance and energy optimized without impacting application KPIs.

Proof-of-concept experimental validation

We validated the enhanced cloud infrastructure by deploying 5G mobile applications of the user plane function (UPF) type. We then generated a realistic workload and applied it to the deployed UPF applications.

All measurements are based on a dual-socket server equipped with Intel Sapphire Rapid CPUs with a power limit of 300W. Only one of the CPUs is used for traffic handling. All figures are based on the CPU socket used for traffic. In our experiments, we measured characteristics when deploying the UPF on our enhanced cloud infrastructure

and compared the results when deployed in a standard Kubernetes cluster. The DPDK (Data Plane Development Kit) based UPF uses a micro-sleep function specially designed to save energy when used in combination with the shim layer [2]. Note that the OS kernel is not involved at all when processing packets.

The shim layer reduces the uncore frequency and scales up the CPU core frequency.

As a traffic model, we used a traffic mix from our internal stability test. We generated the traffic using an external traffic generator that simulates the surrounding network elements.

Proof-of-concept results

Figure 3 shows the processor power consumption, with the medium blue bars demonstrating the power usage when running the unmodified mobile core without the shim layer at idle load (0Mbit/s), at 50 percent of the maximum traffic load and at 100 percent of the maximum traffic load. The first three light blue bars represent the energy consumption when using the energy-optimized mobile core with the shim layer. The dark blue bar to the far left of the figure illustrates the energy usage for best-effort services when deployed on a standard Kubernetes cluster. The light blue bar to the far right highlights the results when using the “boost performance” feature, which utilizes the saved energy to boost CPU performance.

The dark blue bar in Figure 3 indicates that best-effort services, which are not used in traffic and are not real-time critical, consume 177W of energy when deployed using the standard Kubernetes cluster, which is significantly higher than the 104W required when using the energy-optimized mobile core with the shim layer. The reason for this is that the standard Kubernetes deployment cannot differentiate between performance-critical microservices and best-effort microservices, resulting in unnecessarily high energy consumption.

Our results further indicate that processor energy consumption can be reduced by more than 50 percent at idle load when using the shim layer. The energy consumption increases as the traffic load grows but never exceeds the CPU power limit of 300W at maximum traffic load. The energy saved is used to further boost mobile core performance – that is, boost the uncore and CPU core frequency – which provides 28 percent additional performance gain, as illustrated by the light blue bar to the far right of Figure 3. The performance boost can be safely applied within the 300W CPU power budget using the shim layer.

Key benefits

It is clear from our results that the use of separate policies for different classes of workloads enables substantial energy savings. Data plane services are set up to use a rush-to-idle policy and opportunistically enter micro-sleep state in any possible gap between packets. Best-effort services are instead set up to use an energy-efficient policy that does not drive high power by using inefficient turbo-boost frequencies. Together, these policies create a power headroom that can be used to allow higher core frequency for cores that are used for traffic handling. This

enables a higher peak capacity and allows a higher system performance within the same configuration and power envelope.

Processor energy consumption can be reduced by more than 50 percent at idle load.

The energy-saving mechanisms enabled by enhanced cloud infrastructure (especially micro-sleeps) prevent the processors from reaching high temperatures that negatively impact server lifetime. They also prevent performance degradation caused by power limitation at workloads as high as 80 percent processor utilization. Higher peak frequency on the traffic-handling services enables energy-efficient boost peak capacity.

Our measurements also demonstrate that improved power efficiency makes it possible to reduce the amount of hardware by 30 percent while maintaining application KPIs. A smaller system configuration with fewer processors is more energy efficient in and of itself. While the processor power-management mechanisms are efficient in saving energy in the processor, the smaller configuration saves the full energy of the servers.

Conclusion

Our research indicates that it is possible to both save energy and boost performance by implementing an enhanced cloud infrastructure. Access to application knowledge – and

Energy-efficient mobile core networks

application hints – enables the introduction of energy-saving functions in the cloud environment without putting the key performance indicators of applications at risk or requiring modifications to the Kubernetes software. In our proof of concept, we introduced application knowledge in the cloud environment in the form of a shim layer that intercepts and configures the appropriate energy-saving mechanisms. With this approach, we achieved central processing unit energy savings of more than 50 percent at low traffic, a boost in peak performance and a 30 percent reduction in hardware.



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Energy-efficient mobile core networks



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- Ericsson blog, Sustainable networks: Saving energy while maintaining customer experience ↗
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Evolving service management toward intent-driven autonomous networks

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Intent-based autonomous networks entail new capabilities and interaction schemes that impact the use, exposure and management of services. Our goal is to help communication service providers seize the opportunities that service management evolution presents without the risk of disrupting earlier investments and critical business processes.

Service management evolution



The degree of autonomy in a network is measured in terms of the amount of human action that is required to make operational decisions and implement them in the network, with the zero-touch paradigm as the ultimate goal.

The path toward an autonomous network is achieved through the implementation of network operation that is data-driven, self-aware, self-healing, self-optimizing, conflict-resolving, continuously learning, self-adaptive and proactive. An autonomous network is characterized by its ability to:

- Continuously monitor all relevant metrics
- Reason about its own state and identify issues and opportunities to improve
- Automatically find and apply solutions
- Continuously seek to optimize its state
- Detect and mitigate conflicting requirements
- Learn from observation
- Adapt to new situations and demands
- Predict, anticipate and avoid problems.

Ideally, human interaction with an autonomous network should be limited to telling the network what it is expected to achieve and then retreating to monitor that the network is operating as intended. Every situation that requires further human intervention reduces the level of autonomy. These situations are not only the result of errors but can also be

caused by the introduction of new products and services, or by changes to the network, such as upgrading or expanding resources. A highly capable autonomous network would have the ability to adapt itself in these cases.

The concept of intent

The expression and management of operational requirements as separate information objects is an essential step toward autonomous networks. This approach enables a higher level of abstraction as well as separating knowledge about the requirements from the implementation of requirement handling provided by service orchestration and assurance solutions. Until now, the requirements on service characteristics and behavior established and assured in service management have been specified in Service Level Objectives (SLOs). SLOs are life-cycle-managed as artifacts in the service catalog. However, the way SLOs are used in typical implementations of service management with orchestration and assurance is relatively static. SLOs are usually human-determined and human-managed artifacts. Furthermore, changes to the requirements expressed by SLOs do not propagate to existing service and resource instances. They are considered at service creation and configuration time only.

Intent is the next evolution of requirement-bearing knowledge objects [1]. Intent has a dynamic life cycle based on application programming interfaces (APIs) that are specifically designed to allow automated systems to dynamically set and modify requirements as needed at

Terms and abbreviations

API – Application Programming Interface | **CFS** – Customer-Facing Service | **E2E** – End-to-End | **LCM** – Life-Cycle Management
RFS – Resource-Facing Service | **SLO** – Service Level Objective

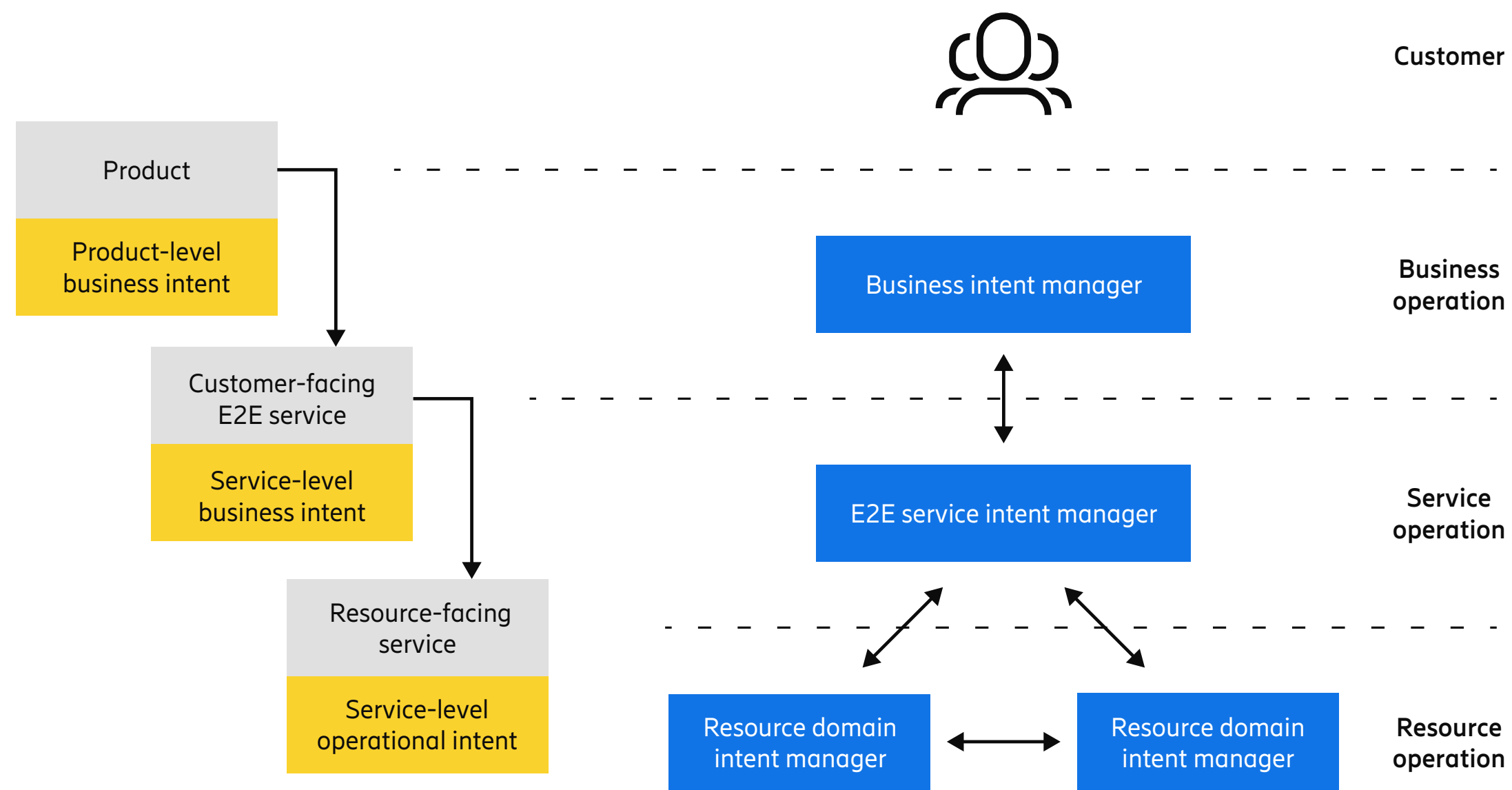


Figure 1: Product and service breakdown in the layered intent management architecture

any time in the life cycle of associated services and derived service instances. A change of requirements expressed by intent is expected to take immediate effect also for compliance of already deployed service and resource instances. This effect on instance level is a key difference between typical SLO and intent-based systems. An SLO bears requirements associated to services, while intent also affects service instances with the expectation that service instances adapt to intent changes. Designing

service orchestration and assurance accordingly is a central challenge in the transition of service management toward intent-based operation.

Intent APIs defined by various standard-defining organizations such as TM Forum and 3GPP (3rd Generation Partnership Project) specify methods to check feasibility, negotiate requirements and provide guarantees. Intent-based operation is also embedded in the generic architecture

Service management evolution

of autonomous networks, in which automated intent managers communicate with each other regarding requirements using intent and provide feedback on compliance to requirements using intent reports.

It is important to note that intent is not necessarily tied to specific services. By providing general requirements for a network domain, intent can indirectly impact how services and products are handled by introducing additional requirements with which they need to comply. However, the recent addition of intent into dedicated service management APIs is the key enabler for the evolution of core commerce processes, and service orchestration and assurance implementations toward intent-driven systems. Intent would complement or replace SLOs as the requirement-bearing artifact. Therefore, the implementation of service management solutions needs to evolve to support the dynamic aspects of intent requirements.

Autonomous network architecture

An autonomous network is subdivided into autonomous domains according to the generic architecture proposed by TM Forum [2]. These autonomous domains are allocated within layers of the network operation software stack, residing on the business, service or resource layer, for example. Within a layer, multiple autonomous domains may exist in parallel or in further subordinate layers.

According to this generic architecture, autonomous domains have a specific responsibility scope within the overall autonomous network. Autonomous domains also interact by setting requirements using intent. This is a common pattern: an autonomous domain has terminal goals that it is required to meet and optimize. To reach its goals, it also needs other domains to contribute to and comply with certain

requirements. Intent is used to communicate and manage these requirements in the various layers in our layered approach to intent management [3].

Every autonomous domain within an autonomous network implements intent management and contains an intent manager, which can be realized using a cognitive loop [4]. Intent managers are the entities that communicate with each other using intent to set requirements. An intent manager also coordinates with other functions within its domain to determine how to achieve the requirements.

Intent managers within autonomous domains are the key entities involved in intent-based operation. Every intent has exactly one owner and one handler. An intent manager plays the role of intent owner if it defines requirements with which another domain needs to comply. If, on the other hand, it receives requirements that it must comply with in its own domain, it assumes the role of intent handler. The right side of **Figure 1** shows the allocation of intent managers in the business, service and resource operation layers.

Intent API evolution

Intent managers use APIs to manage requirements expressed in intents. TM Forum has published the TMF921 API for general purpose intent management. It specifies the communication between intent owners and handlers, enabling operations such as providing intents, reporting on compliance status, modifying requirements and removing intents. It also defines advanced operations such as requesting proposals, providing feedback or checking the feasibility of requirements.

TM Forum is currently using Ericsson's contribution of extensive research and development work in the area

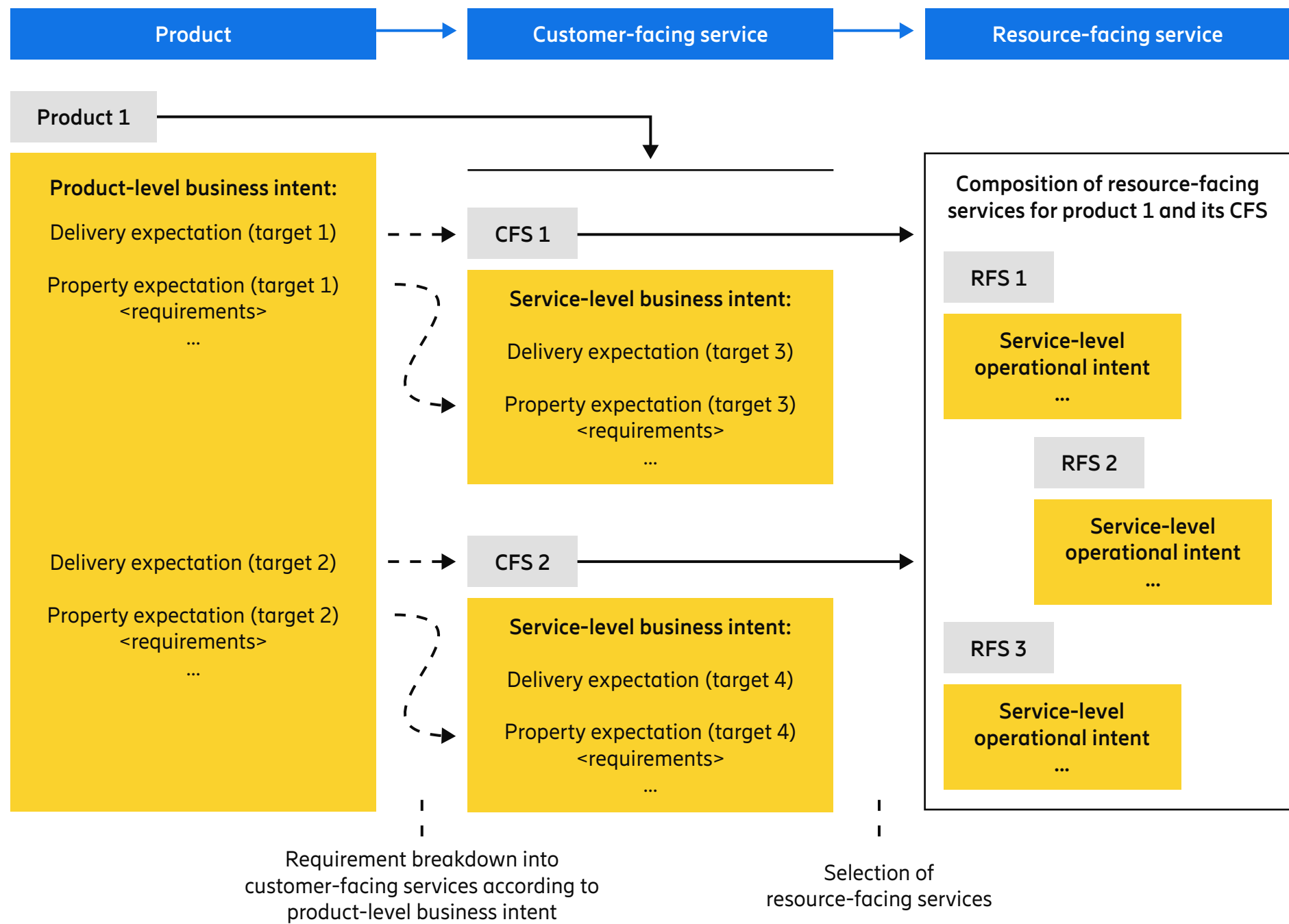


Figure 2: Service referencing from within an intent with example

of autonomous network technologies as the basis for integrating intent management capabilities into its existing APIs for product and service management, as well as its Open Digital Architecture. The API integration process involves several key changes. The first is the addition of intent specifications to the service catalog with links to product and service specifications. The related APIs are TMF620 product catalog management and the upcoming version of service and resource catalog management TMF633 and TMF634.

The second key change is the capture of intent values or references during the ordering phase. These intents are business intents when associated with ordering a product and its associated customer-facing services (CFSs). They will be converted into operational intents when ordering the services used to realize products. Intents are communicated within the product order using TMF622 or service order using the TMF641 API. These APIs will also allow the life-cycle management (LCM) of intent, including dynamic modification and removal.

Service management evolution

The third key change is the creation of a new operation in intent-based systems: intent reporting that creates an implicit feedback channel about the success of handling requirements. Creating reporting expectations will be part of the TMF622 and TMF641 ordering APIs. The intent handler will be responsible for assembling the intent report in line with the reporting expectations and making this information available for external consumption by the intent owner and/or others. The addition of intent qualification to the TMF679 API product offering and the TMF645 API service qualification is another important change, which makes it possible to check the validity of intent against the intent specification. Lastly, changes will be made to make it possible to maintain intents in the product and service registry at the instance level, in relation to the respective product or service instance.

The introduction of intent directly into service management APIs is important for a seamless transition of current business processes and a gradual introduction of intent. Further intent management capabilities will be introduced at a later stage, which will make it possible to explore the feasibility of compliance, negotiate requirements, and both ask for and provide guarantees. This can be realized as part of qualification, service order or through the dedicated TMF921 intent API.

Service management with intent

The service management domain is the starting point for the introduction of intents, as it is the heart of network operation, and intents are a prerequisite of autonomous operation.

The left side of Figure 1 allocates products and services and their respective intents within the layers of an autonomous

DEFINITION OF KEY TERMS

An **intent manager** is a function within an autonomous domain that receives requirements expressed by intent through APIs and coordinates with other functions within its domain to achieve the requirements.

An **intent owner** is an intent manager that defines requirements and thus is the source of an intent.

An **intent handler** is an intent manager that receives an intent and is thus obliged to comply to the requirements within its autonomous domain.

An **intent specification** is a template containing rules and constraints for valid intent content associated with a product or service.

network. The business intent manager is realized by the core commerce function. Business intents will be captured in the core commerce function during product ordering. When the product order is decomposed into service orders, the business intents must also be decomposed and transformed to service intents on multiple levels. The first decomposition step must be applied to the CFS level: these are the services the customer is aware of as part of the ordered product. This is also the level where the classic requirements (phrased as SLOs) would be attached.

Figure 2 provides an example of how the information provided within the intent is used in the successive decomposition process. The end-to-end (E2E) service intent manager will utilize the delivery expectation mechanism. A delivery expectation specifies that something needs to be delivered and provides details relating to it. The intent of a product would specify the set of CFSs to be delivered.

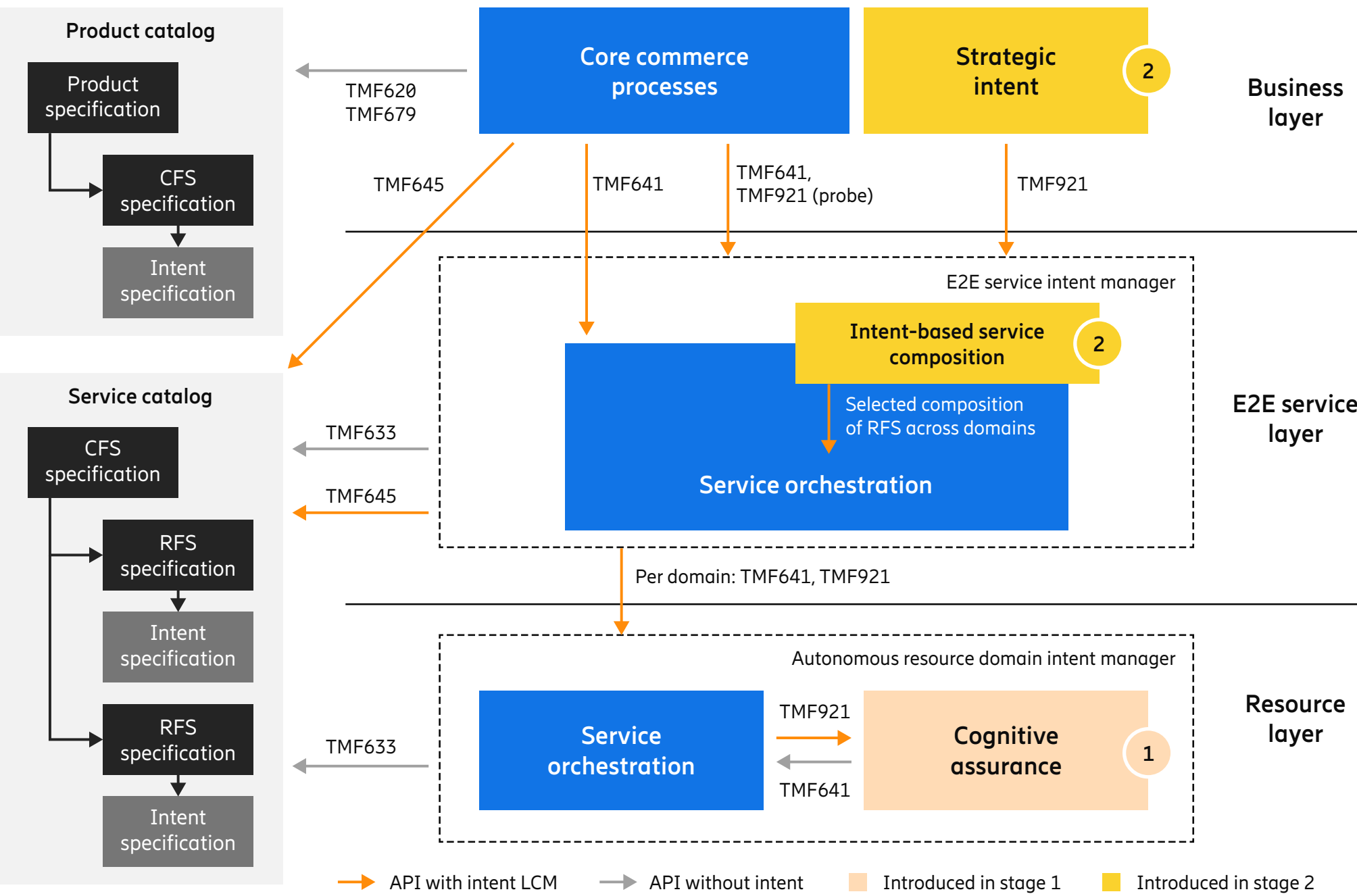


Figure 3: Staged evolution of intent-aware service management

Further requirements stated in the product-level intent, for example, using property expectations and key performance indicator-based conditions, are distributed into the intents for the respective CFSs.

The decomposition of CFSs into applicable resource-facing services (RFSs) with associated operational intents is the responsibility of the E2E service intent manager located in the service operation layer. This intent manager involves functions such as service orchestration.

RFSs are not exposed to the customer. This means the E2E service intent manager has more freedom to compose suitable services. The mechanism of delivery expectations within intents is flexible and can span from specifying a concrete service to be used to loosely categorizing the kind of functionality needed without specifying which concrete service to use. Intent expectations in the CFS would be used to describe properties of suitable RFS choices.

Service management evolution

For example, the service-level business intent associated with a CFS may state that a communication service is needed, providing an E2E latency of 100ms. The E2E service-level intent manager may choose a network-slice-based resource-level service and specify that the radio access network must achieve a latency of 60ms. Consequently, the intent manager selects an RFS matching the needed functionality and breaks down the E2E requirement into a domain-specific contribution.

The receivers of the RFS orders with intent are the resource-domain intent managers. Intents associated with RFSs are referred to as operational intents rather than business intents. Intent managers at this level cater for further translation resulting in domain-specific actions. Examples of this could be provisioning and configuration actions toward network resources or the sending of further intents if resource and network functions are also intent-aware.

Evolutionary introduction of intents into service management

Rather than being disruptive, the introduction of intents into service management should be a smooth evolution in multiple stages. Communication service providers must be able to adjust and upgrade their orchestration and assurance stack without interference to business continuity.

Figure 3 illustrates our proposal for a staged evolution of intent-aware service management. The starting point and base for the evolution is stage 0: SLO-based orchestration and assurance. In this stage, requirements on service characteristics are already separated from service order invocation and captured with SLOs. The service orchestrator uses service-specific policies to analyze and decide how to provision and configure the network considering the SLO-

defined requirements. Artifacts such as TOSCA (Topology and Orchestration Specification for Cloud Applications) models and Business Process Modeling Notation-defined processes guide the orchestrator’s work. The orchestrator also determines the assurance processes needed for automated monitoring and healing.

The introduction of intents should be a smooth evolution in multiple stages.

In stage 0, neither intents nor dynamic LCM of intent requirements involving the possibility to dynamically change the intent content at any point of the service-instance life cycle are used. This is to preserve compatibility with current product and service management processes. Over time, product and service management can evolve in multiple stages to introduce these concepts.

Stage 1: intent-based assurance

Stage 1, shown in the lower part of Figure 3, follows the optional, intent-related extensions to product and service management introduced by TM Forum. Intent is introduced in service operation and replaces SLOs for services using intent-based operation. The business intent details are captured by core commerce processes together with CFS orders utilizing extensions to the TMF641 service-ordering API. This API will make it possible to order services and also manage the respective intent. It is possible but not



mandatory to start capturing business intents when ordering the product. This is already supported by the latest version of the TMF622 product order management API.

Intent specifications are introduced in the product and service catalog and linked to product and service specifications. Intent qualification can be done together with product or service qualification against the intent specification using the TMF679 or TMF645 APIs. Further intent feasibility check and requirement negotiation operations can be performed using the TMF921 intent API or as part of the qualification using TMF645. The intent instance is persisted with the E2E service, and the life cycle of the intent requirements must be managed together with the service.

The E2E service orchestrator would apply a catalog-driven composition to determine the RFSs to be configured and instantiated and would use the intent-defined requirements to make decisions in provisioning. A significant change in stage 1 would be the introduction of an intent-aware assurance loop. The orchestrator will use intents to arm the assurance loop with the requirements to assure using the TMF921 API. These can be the same requirements received in the order or a variation of them based on additional service- and resource-specific concerns to be assured. The related intent reporting will generate notifications to Service Level Agreement management related to the service quality and requirement compliance.

The main task of the intent-based cognitive assurance loop would be to monitor the operated domain and keep it compliant to the operational intent requirements. It would observe through a measurement and analytics infrastructure and act, for example, by reconfiguring resources through

network management or through the orchestrator by modifying services or ordering additional ones. A self-adaptive, intent-driven control loop [4] is well suited for this purpose.

In this stage, advanced features such as decision by utility maximization can be introduced. This concept uses intent to communicate utility information. Utility is a concept specified within intent, which is realized by a utility function that maps an observed metric to a normalized score. This score expresses how valuable the observed value of a metric is. Utility-based decision refers to selecting the most valuable solution alternative with respect to resource availability and currently valid requirements. As such, it also addresses conflict resolution.

As all artifacts, including intent, are optional, the introduction of stage 1 and transition toward intent can be done gradually and on a per use-case and feature basis. For example, intent may be used for new use cases and those services that would profit from dynamic requirement management, while established services continue to be based on SLOs.

Stage 2: intent-based composition and service selection
After intent-based assurance is established, the next stage is the allocation of an intent-handling function as a layer inside the intent manager. Figure 3 shows the allocation of intent-handling functions as a separate process (the selection/ composition of resource-facing services), but in reality, they will be implemented as part of the orchestrator. As a result, the service order with the intent will terminate in the intent management layer. Further decisions are entirely based on the fulfillment of the stated intent requirements.

Service management evolution

If the intent requirements for the new intent are not met, the loop will pick up on it and propose corrective actions. One typical solution in this case can involve the orchestrator and lead to provisioning of resources according to the ordered service. As part of this solution, the intent manager can select the set of RFSs to be used and determine the intent requirement breakdown per subsequent resource domain. This approach can, however, result in no initial action if all requirements expressed by the new intent are already met. The intent manager would then immediately start monitoring continued compliance and would not act until the requirements are no longer being met.

Transition toward intent can be done gradually and on a per use-case and feature basis.

In this embodiment of a service layer intent manager, all operational decisions, including initial fulfillment and subsequent assurance, healing and optimization, are counteractions for fixing the situation that the network is not meeting all its requirements in the most optimal way. These actions can involve the configuration of resources, selection and orchestration of RFSs or the breakdown of requirements received in a CFS intent into resource-domain-specific intent. This means, the intent-based service composition provides a flexibility layer based on the dynamic composition of services, with continuous monitoring and optimization.

This concept also introduces a strong separation of concerns and decoupling of system layers, as the resources used to implement a customer requirement and deliver the services a customer expects would not be directly implied by the services or products being ordered by the customer. Instead, the autonomous service layer would find solutions based on the available RFSs to optimally utilize resources. This also means that the service management system would not be bound to inflexible vertical integration of predefined and coupled artifacts. This is an environment in which methods based on artificial intelligence, with its potential to find creative solutions and optimize beyond human capabilities, can deliver great value.

Stage 3: adapting existing resource-facing service instances
The intent-based service composition function introduced in stage 2 uses RFSs flexibly, but still chooses from a catalog of predefined services. In stage 3, the idea is that the service instances themselves can evolve. The autonomous network could introduce new service management artifacts or variate existing ones to better represent available resources. Furthermore, changes to the resource usage and configuration can be implemented through modification of service instances. At this stage, the autonomous network has full adaptation capabilities and can reshape existing service instances in line with new requirements involving new service specifications.

For example, new resources may be supplied by a new business partner. They may also provide better scaling capabilities or better response times or be more efficient from a total cost of ownership point of view. These new resources will be made available to the orchestration function by adding new resource specifications and



associated RFS specifications. New service orders can utilize them immediately, but it would also be beneficial to replace already provisioned resources and service instances.

In stage 3 of the evolution, an autonomous network can decide to reshape existing service instances and eventually transition them to the new resources, once they are made available. The decision to transition service instances to new resources would be an optimization decision taken by the intent-handling function, which is continuously producing new solution alternatives to improve the system state and raise business value. This process would assess that a dynamic re-composition action will lead to a higher level of intent fulfilment and business utility. In other words, once the new resources are available, the intent-management function will identify the optimization potential and execute the required actions to achieve it. This involves the recomposition of the service by the service management system. This type of dynamic self-adaptation is a key capability for autonomous networks.

Conclusion

While it is clear that the creation of highly capable, zero-touch autonomous networks will require major changes in network operation and service management, our research demonstrates that it is possible to introduce the necessary changes gradually with increasing levels of autonomy over time. Self-adaptive capabilities are often challenging to achieve and require the use of artificial intelligence methods in combination with interfaces and architectures that strengthen abstraction and separation of concerns between the layers of network operation. Service management provides an excellent abstraction of resources for this purpose, especially if combined with dynamic requirement management of intent-based interactions.

Ericsson’s staged approach to service management evolution enables communication service providers to take the journey at their own pace suited to their own market situation, business model and strategy. We recommend the introduction of intent-based operation only for those services and network domains that would currently profit from flexible requirement handling. Other use cases can remain as they are. In the first stage, it is possible to gain advantages from autonomous networks technology without making changes to existing service management processes. Most existing service management artifacts and processes can be reused in intent-based systems until there is a clear business case for further evolution.

The authors



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Service management evolution



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Further reading

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- Ericsson blog, Want to get ahead in service management and orchestration? ↗
- Ericsson, E2E service orchestration ↗



CTO Technology Trends 2024: Key trends shaping the future of digitalization

Author:
Erik Ekudden, CTO

This year marks the 10th anniversary of Ericsson’s CTO tech trends article series, making this an opportune moment to reflect on how the trends have evolved over time. From the beginning, the focus of these articles has been on anticipated technology evolution, gradually encompassing driving use cases and, in recent years, enriching the use cases with business innovation.

In the 2019 edition of the CTO tech trends article, I outlined my vision for the network platform of the future, emphasizing the crucial role of consistent and open business interfaces – that is, application programming interfaces (APIs) – that offer a multitude of capabilities to all users. Since then, we have catalyzed a shift in the industry, placing this vision at the forefront not only in the mobile sector but also among partners, enterprises and developers.

As a result, we are now on the cusp of a massive transformation as the telecom industry transitions toward a platform economy. The mobile industry is today dominated by smartphones and best-effort mobile broadband (MBB) services in a mature market, and service innovation is a top priority on the industry agenda to spur profitable growth by utilizing new business models and more advanced network characteristics.

As we embark on this transformative journey, our overarching objectives remain clear: to ensure the cost efficiency, sustainability and trustworthiness of high-performing programmable networks. Ericsson has prepared a toolbox, equipped with a developer platform, network APIs, network exposure and network slicing to enable differentiated services. Equipped with these powerful tools, we stand ready to shape the next era of wireless connectivity and business innovation.

Analyzing the evolving needs of society and business

Both the business community and society at large are continuously searching for more efficient and sustainable solutions with the potential to improve profitability, provide economic stability and stimulate growth. Today, many of these solutions are enabled by digitalization, electrification and automation. The electrification of everything includes the energy transition toward renewable energy sources such as solar cells, distributed energy storage and the transition from combustion to electrical engines. Advances in automation include autonomous operations and artificial intelligence (AI) powered learning, reasoning, problem-solving and language understanding, with generative AI playing a significant role. Generative AI not only boosts productivity in development processes but also plays a pivotal role in fostering innovation. Examples of this include efficient data analysis that provides insights and predictions, the personalization of experiences and new methods of delivering products and services.

The digitalization of society and businesses is advancing rapidly in tandem with the evolution of connectivity, AI, and compute and cloud-infrastructure technologies. In the case of 5G networks, these technologies are used to facilitate programmable and differentiated networks for more efficient service delivery. The performance, integrity and versatility of the global 5G network serve as a powerful digitalization catalyst.

Ericsson’s mission is to lead our industry in creating a stronger, more vibrant and more innovative business landscape. We are also driving cross-industry collaboration to form an open network ecosystem with



DIFFERENTIATED NETWORKS – A FEW EXAMPLES

Sony: 5G-enabled media production through an API
Using the existing 5G network for live broadcasting means reduced setup time, fewer cables, lower costs and unprecedented mobility during broadcast production – with no need to compromise on quality and latency. This is ideal for everything from high-profile news and public events to live sports and TV studios.

Toyota: enhancing the driving experience with 5G
Integrating high-performing connectivity into vehicles benefits both Toyota and its customers: owners get enhanced services that make driving safer and more pleasant, and Toyota gets more reliable vehicle insights and seamless product updates. The use cases for connected mobility include infotainment and navigation, vehicle teleoperation and over-the-air updates.

Telia NorthStar: advancing 5G with developers for tomorrow’s use cases
At the center of NorthStar is a purpose-built innovation network, with a new 5G core connected to Telia’s existing network. In this experimental environment, new use cases will be developed across industries – from differentiated service solutions that provide superior quality in Teams meetings, to ultralow latency connections that enable remote-controlled trucks in mining operations.



leading industry players from devices and chipsets to the cloud providers and application developers. The global network platform we envision will be easily accessible to any enterprise or developer, and will address society and business needs through viable business solutions that spur innovation, collaboration, competition and investments.

The challenge of meeting differentiated needs

Today’s consumer, enterprise and developer applications – as well as those of the future – comprise a vast array of use cases with very different network needs. These network needs range from low resource-utilization requirements, such as text messaging, to more resource-intensive needs, like immersive communication experiences. Requirements on throughput, latency and other network characteristics to optimize the user experience vary widely.

For instance, consider the contrast between video streaming and immersive communication. Video streaming employs buffering to ensure a smooth user experience, accommodating potential network fluctuations. On the other hand, immersive communication relies on near-real-time interactivity to provide a seamless experience, allowing users to interact with virtual objects, physical environments and other users.

In addition to variations in terms of their network resource demands, different segments, applications and use cases also have widely varying requirements when it comes to reliability, coverage, security, identification and authentication. For example:

- Consistent and reliable connectivity is crucial for government services, such as public safety operations.



- Specific uplink characteristics are necessary for interactive immersive communication.
- Enterprise control over all assets, achieved through autonomous identification and authentication, as well as a secure access service edge, is imperative.
- Comprehensive outdoor and indoor coverage, along with advanced positioning capabilities, is essential for enabling features like 3D mapping support and location checks for fraud protection.

The high-performing differentiated network will address these various needs in a cost-effective and resource-efficient manner. Given that high performance is not always a requisite, optimal efficiency can be achieved through the use of orchestration techniques that dynamically adjust network resources to meet actual needs, while simultaneously ensuring the required quality of service (QoS) for the different user groups. This underscores the importance of future networks being not only open but also programmable.

The high-performing differentiated network approach caters to new segmented and dynamic business models spanning application, device, network and cloud. While today’s best-effort subscription will continue as a basic service offering, the differentiated approach will enhance the value of network offerings and subsequently increase the willingness to pay.

Four trends driving the evolution of high-performing programmable networks

The pace of mobile innovation over the past 30 years has been extraordinary. For instance, we have achieved a 100-fold increase in data speed every decade, amounting to a one-million-fold leap over 30 years – surpassing the growth projected by Moore’s law. In the next decade, we anticipate another 100-fold improvement. This will be driven by the combination of high-performing programmable networks, service differentiation underpinned by AI-enhanced network automation and groundbreaking technological innovations in both software and hardware. Together, these elements will lead to a new era of connectivity and business innovation.

TREND #1:
Using network programmability to meet the need for service differentiation

Network programmability fundamentally revolves around two key aspects. The first is the programmability of the network infrastructure, which is managed by the communication service provider (CSP) to efficiently utilize and operate the network resources. The second aspect is on-demand service programmability, which allows applications to dynamically request network resources using network APIs for various service levels. The network APIs are accessible through a network infrastructure exposure layer to network aggregators, such as Vonage, which offers globally available network APIs and services to the ecosystem of application service providers and application developers.

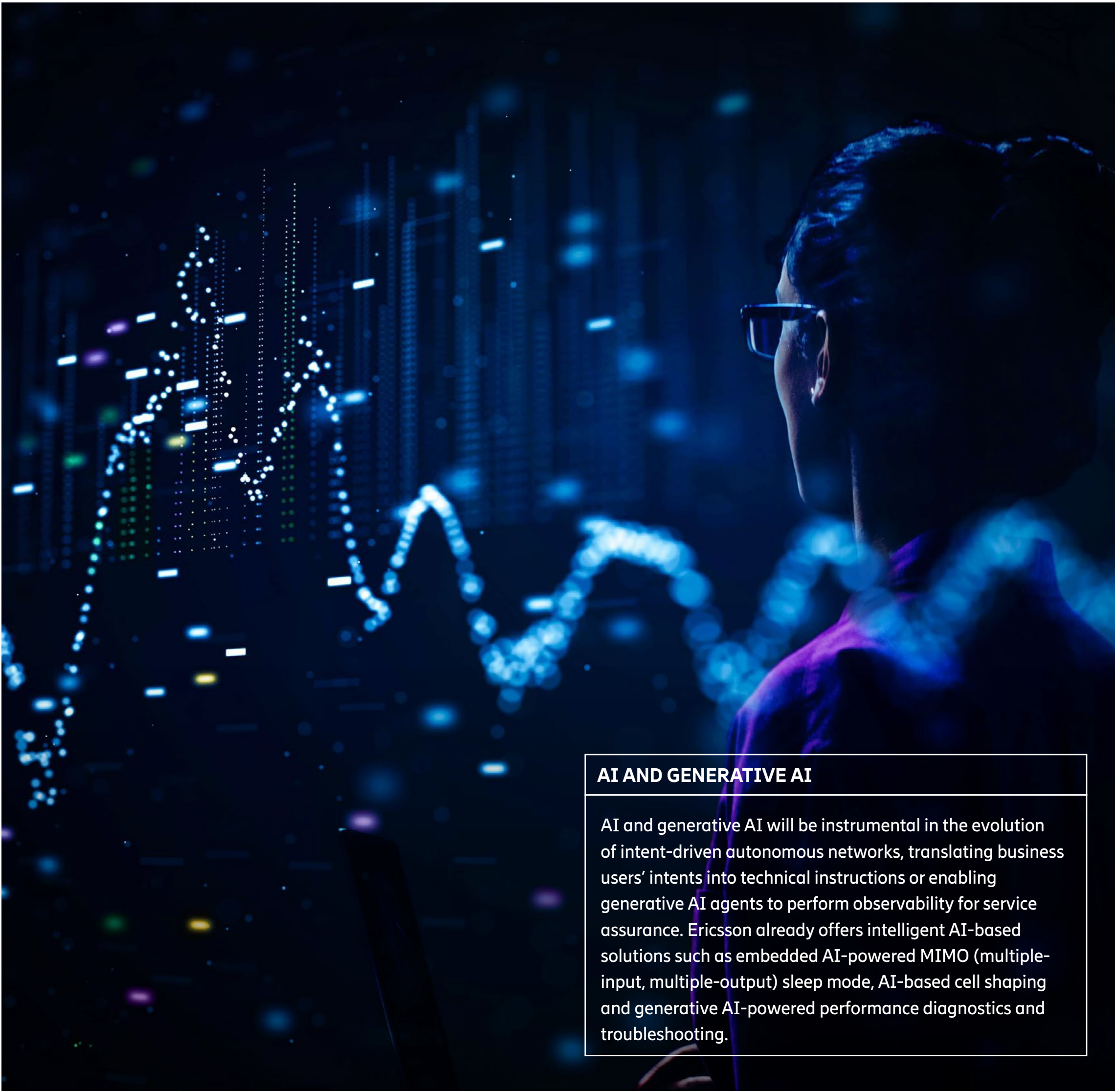
The CSP exercises control over the network resources, verifying availability. This process is facilitated by the

management system of the programmable network, which employs AI-based observability agents and mechanisms. These agents enable the system to automatically respond in real time as well as anticipating future needs when required, thereby ensuring optimal service provisioning and network operations utilization.

The interplay of network infrastructure programmability and network APIs is fundamental in catering to applications and services that require both diverse and dynamic network performance metrics. Hence, programmable networks will offer predefined performance characteristics that range from best effort to guaranteed, thereby meeting the array of connectivity needs. These predefined performance characteristics can be structured into four main classes:

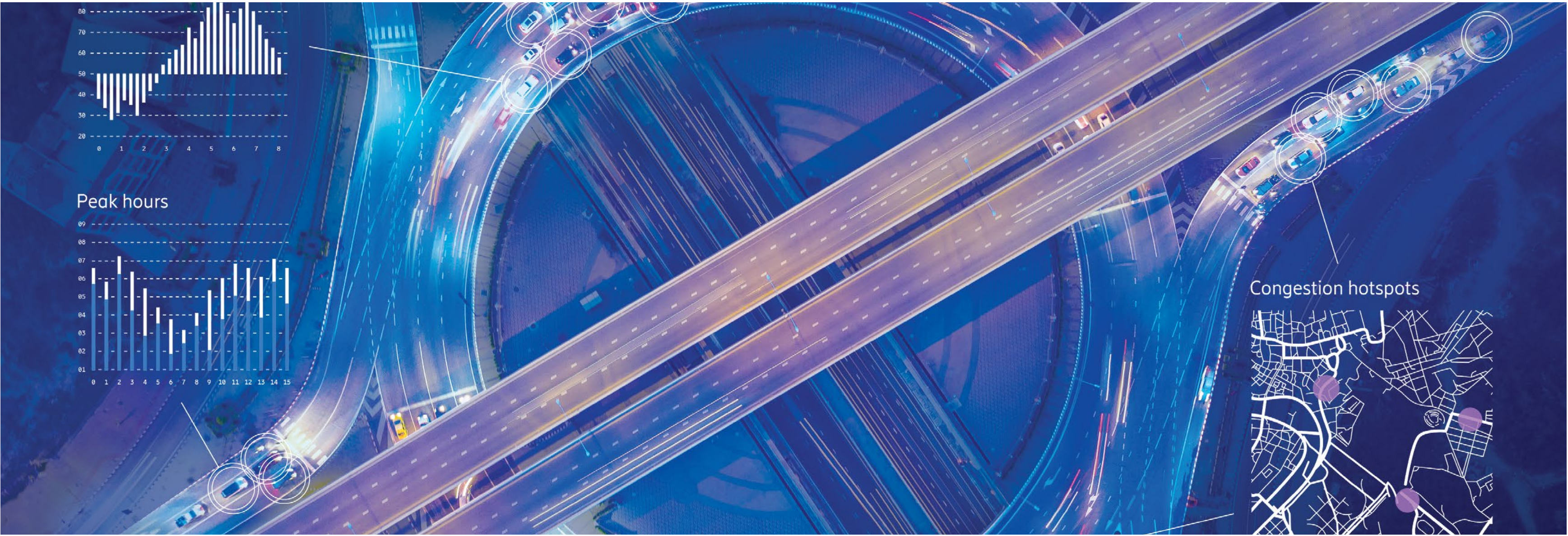
- Class 1: Adaptive Buffered – best-effort throughput and best-effort latency that caters for applications developed for traditional MBB.
- Class 2: Fixed Buffered – guaranteed throughput and best-effort latency optimized for use cases like background application communication, such as over-the-air upgrades.
- Class 3: Fixed Immediate – best-effort throughput and guaranteed latency optimized for applications that require consistent response times, like uninterrupted live video feeds or video calls with adaptable resolution.
- Class 4: Adaptive Immediate – guaranteed throughput and guaranteed latency optimized for applications that demand high reliability and predictability, such as live broadcasting or the remote control of vehicles.

Guaranteed performance is subject to coverage and can be provided with different levels of performance metrics, offering further degrees of differentiation and customization



AI AND GENERATIVE AI

AI and generative AI will be instrumental in the evolution of intent-driven autonomous networks, translating business users’ intents into technical instructions or enabling generative AI agents to perform observability for service assurance. Ericsson already offers intelligent AI-based solutions such as embedded AI-powered MIMO (multiple-input, multiple-output) sleep mode, AI-based cell shaping and generative AI-powered performance diagnostics and troubleshooting.



based on specific application needs. The programmable network infrastructure will therefore provide advanced observability with high granularity and dynamism through mechanisms such as network slicing.

Gaming is an excellent example of a use case that utilizes all four performance-characteristic classes. Class 1 is sufficient for tasks such as downloading menus, while class 2 is best suited to receiving new maps/terrains, for example. Class 3 is the optimal choice for actions in competitive multiplayer games, while class 4 is ideal for tasks such as receiving remote-rendered screens.

The required guaranteed performance levels are requested by the applications through the network APIs, which enables

the CSP to charge for the guaranteed network performance provided. The guaranteed performance can be purchased in different ways, such as a one-time purchase triggered by the application or a consumption-based purchase bundled with the application.

In summary, programmable networks enable service differentiation for consumers and enterprises by providing guaranteed QoS. This not only ensures robust and reliable connectivity but also empowers CSPs to monetize network services and capabilities, including Service Level Agreement assurance, observability and network APIs. This will enable CSPs to compete on both customization and performance.

TREND #2:
Leveraging telecom AI to boost the high-performing programmable network

The high-performing programmable network will evolve into an intent-driven, autonomous system, capable of supporting diverse service needs with optimum cost efficiency and sustainability. This transformation will be powered by AI, which will drive performance optimizations across business, service and resource network operations. The solution entails a hierarchical structure of embedded AI in core networks (CNs) and radio access networks (RANs), AI-based management and orchestration functions and generative AI-based network operation. This paves the way for autonomous network operations and optimization

of capabilities such as energy performance, capacity, coverage, throughput and resilience. Humans maintain control, providing system requirements and overseeing their fulfillment, by expressing intents rather than dictating specific actions.

AI-powered execution environments must be available everywhere, and AI training could be colocated if needed to support the increasing usage of data for multiple purposes. The distributed data infrastructure is managed through a common data plane to enable efficient life-cycle management, transport and storage of network data. The life-cycle management relies on a fully automated trustworthy model that constantly improves and follows data changes, to achieve system-wide end-to-end gains, such as constantly improved resource efficiency. The data management of the network includes control of when and where data will be consumed to comply with privacy and other legal constraints. The network therefore provides a flexible data observability adaptable to the requirements of the data consumer.

The distributed AI-powered execution environments will be capable of handling any AI model, including generative models. These AI models can be distributed across different locations, training collaboratively to protect data privacy and utilize resources more efficiently. In 6G, we aim for telecom AI-native solutions including the air interface, context-aware sensing and positioning. Today's embedded AI functions in CNs and RANs will be reinforced with generative AI capabilities.

Standardization efforts for distributed AI and data in networks are underway in the 3GPP (3rd Generation Partnership Project), while intent-capable network interface



standardization is progressing in TM Forum, O-RAN and the 3GPP. Ericsson is integrating trustworthy AI principles into the development process by following the EU guidelines for trustworthy AI. Overall, these developments contribute to an AI-native architecture that enables innovative network functions and services through distributed AI and data, transforming the network into a platform for innovation.

TREND #3:
Transitioning toward a network architecture with horizontal layers to enable innovation for enhanced and expanded capabilities

For the time frame of 2028, our development organization is preparing for the commercialization of evolved 5G features and capabilities including intelligent RAN automation, RAN features that enhance power efficiency, and time-critical communication that enhances applications such as cloud gaming, industrial automation, the Internet of Things, extended reality and emergency services. Further, the usage of 5G access technologies is expected to extend into non-terrestrial networks (NTNs) and digital airspace during this time frame, starting with the first demonstrations of 5G enabling communication services between NTNs and smartphones. Many of these evolved 5G features will serve as templates for the fundamental 6G design choices.

Perhaps most importantly, during this time frame the network implementation will be restructured into horizontal layers. Each layer will be developed and specified according to its own pace and merits, while retaining the possibility to scale up for global adoption. This includes service-based CN and cloud RAN solutions both fully utilizing cloud-native and AI technologies. The architecture also includes open interfaces such as the air interface (New Radio) and



the fronthaul interface (Next Generation – Lower Layer Split). The open network interfaces provide an opportunity for network modules to be developed separately, while at the same time being interoperable with other modules in the network. The horizontal architecture provides a more flexible, adaptable and customizable network to address the broad range of needs and requirements.

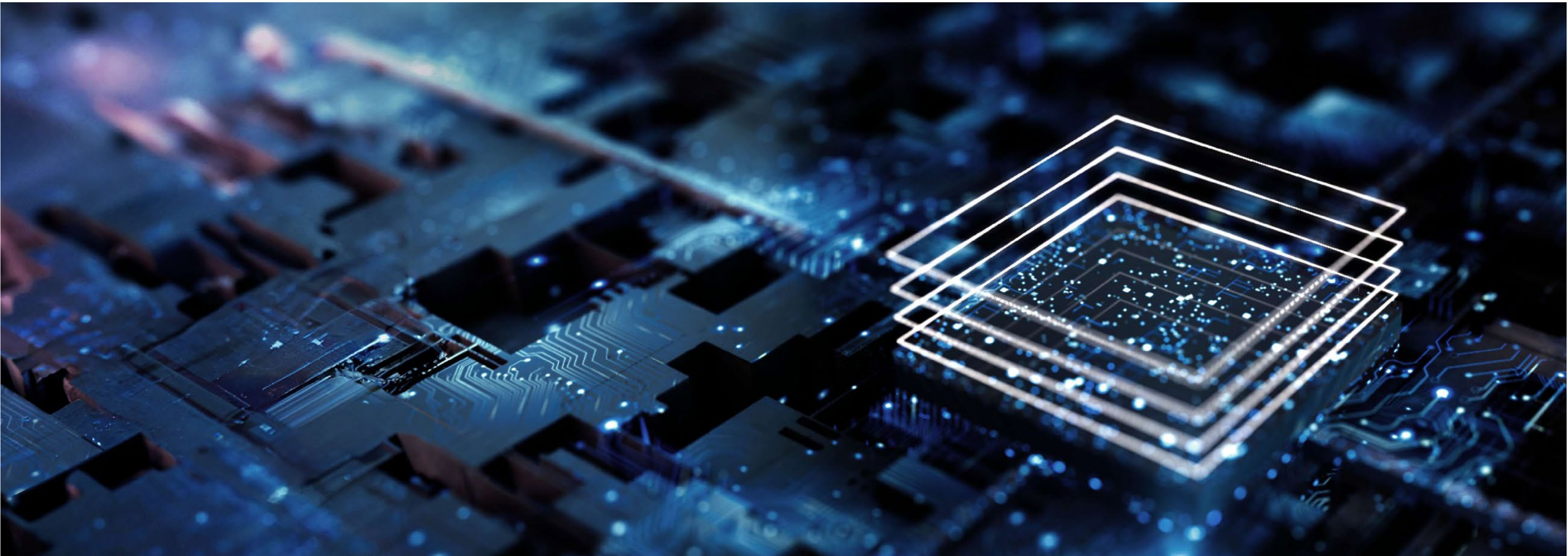
Entering the next decade, the 6G network will bring both enhanced and expanded capabilities. Built on evolved 5G, it will have unique capabilities including wider spectrum range, extended coverage, faster data speeds and lower latency for increased network capacity. Examples of future 6G technologies include full duplex radio solutions and larger massive MIMO arrays. These solutions also enable reuse of the current site grid for the new centimeter-spectrum band

as well as coverage and capacity improvements in the uplink. Further, the 6G network will provide enhanced mobility with increased relative speed between devices and radio base stations, both on the ground and in the air. Crucially, 6G will be built according to a lean design approach that optimizes energy efficiency by integrating zero-energy devices and delivering fully dynamic power management that enables scalable power consumption adapted to the network traffic.

Expanded capabilities encompass integrated radio-based sensing utilizing radio transmission to detect radio-wave propagation in the physical environment. Sensing is enabled by the reuse of higher frequency and wider spectrum bandwidth as well as the infrastructure of the 6G network. The resulting sensing data includes accurate three-dimensional and real-time information about the

physical world to enable applications such as digital twins and spatial maps. Integrated sensing and communication (ISAC) allows for synergies between communication and sensing functions, where the data obtained from sensing can enhance communication (through adaptive beamforming based on environment mapping, for example) and vice versa.

The 6G network will be capable of handling dynamic and temporary deployment needs that address the versatility of new use cases or specific needs, making it possible to support flexible levels of resilience and reliability, for example. Another example of this is the ability to meet specific deployment requirements such as on-premises processing, interworking with other systems and the integration of temporal or mobile nodes.



Finally, trustworthiness, resilience and availability are all essential characteristics of a secure, high-performing network as well. At Ericsson, we are using zero-trust principles to develop a zero-trust architecture (ZTA) that secures network resources and data across all elements, including identity, credentials and access management, as well as end points and their interconnecting infrastructure. The ZTA integrates security components, workflow orchestration and access policies that strictly adhere to these principles with the objective to protect all data and resources, regardless of whether they are located on premises or in the cloud.

TREND #4:
Pursuing innovations over the next decade that enhance performance, sustainability and security

Novel technologies that make it possible to achieve both high performance and energy efficiency, including the diversification and specialization in hardware, are an important research priority since mainstream processors can no longer be relied upon to deliver the performance improvements that Moore’s law dictates. New processor architectures, such as neuromorphic and quantum processors, are expected to be available for commercial use for specific use cases within the next 10 years. Further, integrated photonic interconnects will increase data rates,

significantly reduce energy consumption and enhance flexibility in resource utilization to meet the growing demand for high-speed data communication in future networks.

The slowing down of Moore’s law will also drive requirements on more performant real-time systems-level programming. To innovate, software developers and system architects will need to develop lower-layer software and will therefore become more adept at exploiting the capabilities in the new processor and memory architectures. Besides real-time performance, increased focus will be put on capabilities such as security and availability in hardware-near software. One example is low-level programming languages such as Rust, which offer both memory safety

and fine-grained control over system resources: two features that are increasingly important for writing performant code without sacrificing safety. In the case of AI, the focus will be on prioritizing software and algorithmic innovation, utilizing specialized hardware, and the efficient use of available computational resources.

Conclusion

Ericsson is committed to building high-performing programmable networks that stimulate growth and innovation by exposing network assets through application programming interfaces that are accessible to developer platforms. This approach offers the advantage of catering to a diverse range of application needs, from best-effort to guaranteed connectivity, and supporting performance-based business models to deliver optimal network performance at the right cost.

The combination of 6G and AI is fundamental to this journey. I often explain it this way: the 6G access network is the muscle with unlimited reach that will power a vast array of applications and services, providing differentiated, robust and high-speed connectivity. Meanwhile, the 6G embedded network AI is the distributed brain of the system, bringing adaptability and efficiency to 6G, controlling and optimizing its full operation.

As we lead the way into the next era, Ericsson aims to spur innovations that enhance the lives of people around the world, make businesses more prosperous and foster long-term, sustainable economic growth. My colleagues and I could not be more excited about the plethora of opportunities that lie ahead. Whatever your role in the ecosystem, we welcome you to join us on the journey.



The author



As Group CTO, Erik Ekudden is responsible for Group Strategy and Technology. His extensive experience of working with technology leadership globally influences the company’s strategic decisions and its investments in 5G, 6G, edge computing, AI, augmented/virtual reality and the Internet of Things. Ekudden’s leadership builds on his decades-long career in technology strategies and industry activities. He joined Ericsson in 1993 and has held various management positions in the company, including Head of Technology Strategy, Chief Technology Officer Americas in Santa Clara (US), and Head of Standardization and Industry. He is also a member of the Royal Swedish Academy of Engineering Sciences and the publisher of Ericsson Technology Review.

Further reading

- [Ericsson Technology Review, Beyond bit-pipes – new opportunities on the 6G platform ↗](#)
- [Ericsson Technology Review, Service quality monitoring – an essential tool in the digital economy ↗](#)
- [Ericsson Technology Review, Evolving service management toward intent-driven autonomous networks ↗](#)
- [Ericsson white paper, Co-creating a cyber physical world ↗](#)
- [Ericsson white paper, Zero Trust Architecture for advancing mobile network security operations ↗](#)
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Cloud-native observability of telecom applications

Authors:
Jörg Aelken, Johan Wallin, Tadhg Deasy, Vincent Desbois, Magnus Standar

Solutions that provide observability for cloud-native telecom applications must be able to handle the dynamic, decoupled and highly distributed nature of telecommunication application components in the cloud-native environment. This requires new approaches to collecting, storing and processing telemetry data using and advancing the de facto standard technologies from the cloud-native open-source ecosystem.

Cloud-native application observability



Application observability is a critical capability in network, system and service management, providing insights into an application’s current state and supporting decision-making about changes or updates required to meet operational requirements.

The telemetry data exposed by an application is used as input to many management functions, such as assurance and optimization, allowing the operator to effectively maintain the network. Telemetry data is equally vital to developers, providing essential insights for runtime troubleshooting as well as in the transition to DevOps.

Since its inception, the telecom industry has always utilized a highly distributed architecture, even when capability was delivered in the form of fixed, integrated appliances. As telecom applications evolve from these fixed appliances to be deployed as cloud-native applications on independently provisioned platforms and infrastructure, it is crucial that observability methods and technologies also evolve to enable effective management of the applications, including their provided network functions. The desired flexibility and dynamic scaling of cloud-native applications mean that existing observability patterns will not be sufficient in terms of timely access to data nor in supporting the correlation of data across a disaggregated deployment.

Terms and abbreviations

API – Application Programming Interface | CaaS – Container as a Service | CNCF – Cloud Native Computing Foundation | CSP – Communication Service Provider | eBPF – Extended Berkeley Packet Filter | IaaS – Infrastructure as a Service | K8s – Kubernetes | OSS – Operations Support Systems | OTel – OpenTelemetry | OTLP – OpenTelemetry Protocol | PaaS – Platform as a Service | RED – Rate, Errors, Duration | SDK – Software Development Kit | USE – Utilization, Saturation, Errors

In the face of this new, disaggregated deployment model for cloud-native applications, the generation, collection, persistence and post-processing of telemetry data must change to ensure that cloud-native applications are sufficiently observable. This includes the capability to correlate telemetry data from the applications and the underlying platforms and infrastructure. To ensure that data volumes are manageable, applications must expose appropriate volumes of telemetry data to support the wanted use cases, with the capability to contextualize the overall network deployed in an external higher layer management system.

Two sets of challenges to overcome

Cloud-native applications are designed so that their parts can be developed, deployed and scaled independently of each other [1]. These architecture and implementation choices are selected to enhance the speed of development and to achieve high system elasticity. However, these objectives are threatened by the high complexity of cloud-native applications and bring new challenges to the telecommunications industry. Any team that is responsible for ensuring the reliability of services that cloud-native applications provide must have the ability to handle:

1. **Complex requests and operations that traverse a high number of deployed components**
2. **A multitude of moving parts due to independent life-cycle handling (some deployed components versions may coexist for the first time in production)**

- 3. Dynamic scale-out/in operations that change the dependencies of deployed components in runtime
- 4. Disruptive effects on the application caused by faults that originate from the highly layered and separately managed cloud infrastructure
- 5. An increase in reported problems that are unexpected and/or have unknown root causes
- 6. A huge amount of data to digest.

On top of these six generic technology-transformation challenges, communication service providers (CSPs) face three additional telecom-specific challenges. Traditional telecom applications are typically realized using well integrated nodes that provide standardized network functions, which expose telemetry data that describes the state of the functions provided. This observability follows traditional telecom management paradigms as specified in the ITU-T (International Telecommunication Union Telecommunication Standardization Sector) M.3100 Generic Network Information Model [2]. Therefore, the implementation of cloud-native telecom applications requires the ability to:

- 1. Collect enough telemetry data over costly transmission links from highly distributed systems
- 2. Efficiently monitor and troubleshoot an application in scenarios where trust and responsibilities may be shared between multiple business actors (the application owner, the cloud provider and Ericsson as the software vendor)
- 3. Continue to provide a functional view while simultaneously keeping the flexibility and detailed observability provided by the cloud-native approach.

The path toward cloud-native telecom application observability

According to the Cloud Native Computing Foundation (CNCF), “Observability is a system property that defines the degree to which the system can generate actionable insights. It allows users to understand a system’s state from these external outputs and take (corrective) action” [3].

The CNCF definition has multiple implications. On the one hand, telemetry data must be collected from all parts of a system. On the other hand, telemetry data must be processed and provided to users, both humans and machines, to answer their many questions. The answers serve overall observability goals such as problem root-cause analysis, closed-loop enforcement, monitoring and so on.

The CNCF’s definition also hints at different maturity levels in observability solutions. While it is important to detect when an application is under predictable conditions (the knowns), we also need a way to deal with what we do not understand and what we did not expect (the unknowns). This ambition level is connected to an organization’s maturity and is equally relevant to its operational and software design business processes. **Figure 1** illustrates our model for observability maturity.

- Telemetry data requirements**
- According to the cloud-native industry, the three pillars of observability are:
- 1. Metrics
 - 2. Distributed traces
 - 3. Logs.

Cloud-native application observability

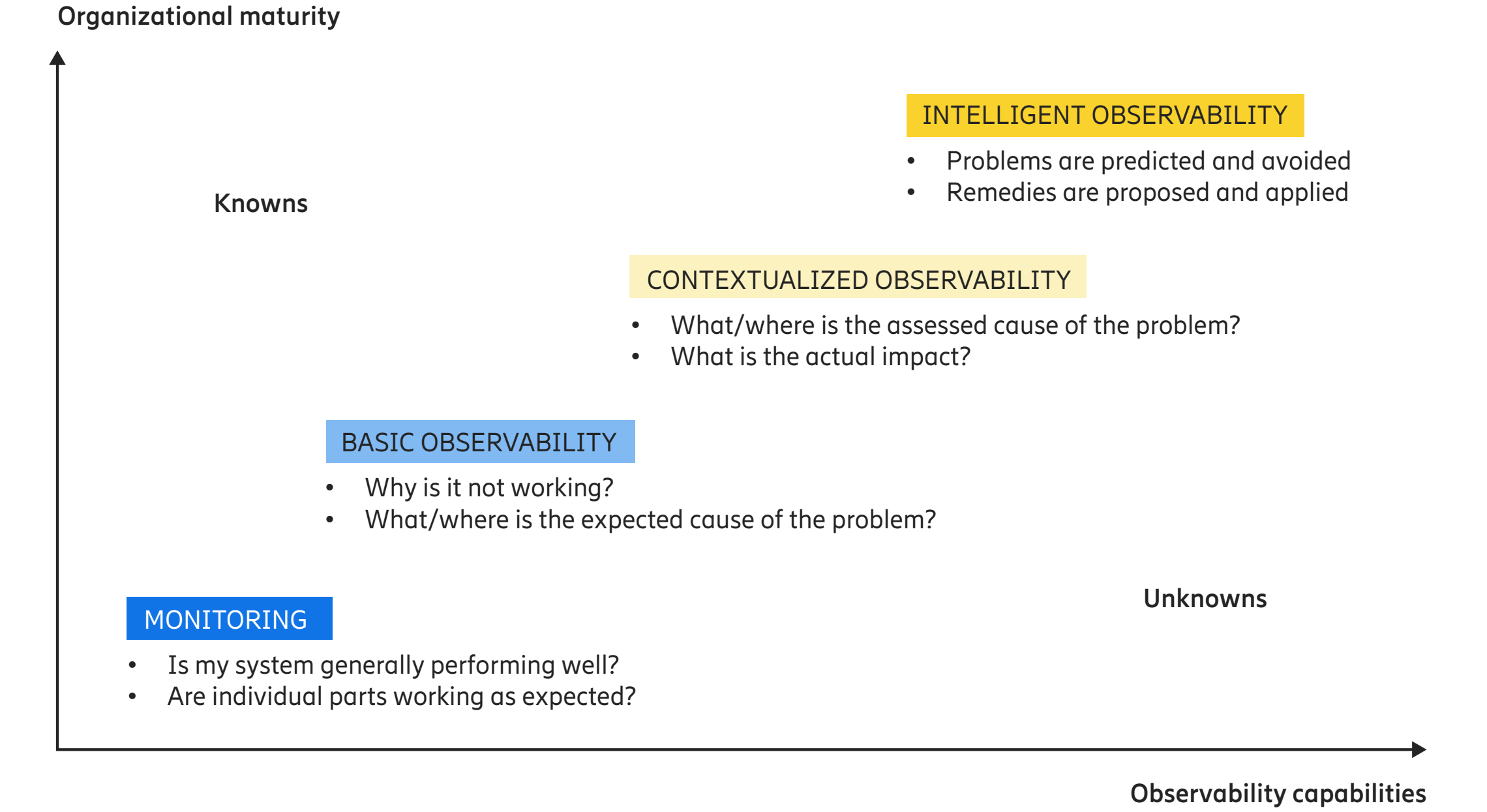


Figure 1: Observability maturity model

Combined telemetry data types based on these three pillars can significantly boost the maturity of an observability solution, provided this data is carefully designed and collected throughout the application, its platform and underlying infrastructure.

From Ericsson’s perspective, there are two indispensable factors for success: the adoption of capable telemetry frameworks and the technical quality of the instrumented

telemetry. The adoption of capable telemetry frameworks encompasses instrumentation libraries and telemetry collector components, providing key enablers such as:

- Data models that enable structured telemetry data
- Aligned time stamps (including synchronized time across highly distributed telecom networks)
- Aligned identifiers (resource identifiers, request/operation context identifiers)



Layer	Telemetry data types			
	Metrics	Distributed traces	Logs	Other types
Application functional layer	Function-centric app custom metrics	n/a*	- Functional logs - Security and audit logs	App business logic events
Application software layer	Software-centric app custom metrics	App distributed traces	App container logs	Continuous profiles
PaaS	PaaS services custom metrics	PaaS services distributed traces	PaaS services logs	n/a
Caas	- K8s state metrics - K8s resource metrics - K8s control plane custom metrics	n/a	- K8s control plane logs - K8s audit logs	- K8s events - eBPF traces
IaaS	Host resource metrics	n/a	Host system logs	n/a

* Distributed traces for the application functional layer are captured in application-domain-specific and other types of business logic events.

Figure 2: Telemetry data types across the cloud-native horizontal layers

- Efficient streaming protocols, such as gRPC (Google Remote Procedure Call).

The technical quality of the instrumented telemetry can be assured through the use of complementary methods such as the four golden signals [4], RED (rate, errors, duration) [5] and USE (utilization, saturation, errors) [6], which set expectations on the scope of exposed metrics to achieve a decent observability level.

It is important to realize that the quest for high observability maturity does not end with the three pillars of metrics, distributed traces and logs. High observability maturity

requires a multilayered perspective, as shown in the table in Figure 2. For example, to understand unexpected application performance problems, you must collect continuous profiles. To understand the root cause of Kubernetes (K8s) runtime failures or pod scheduling failures, you must collect K8s events deeper down in the stack.

The end of node-centric management

In the telecom domain, the integrated node has traditionally been the source of everything: the aggregated set of hardware, the singleton software package, the entry point for configuration, the resulting network function service and the emitter of the aggregated telemetry for all contained

Cloud-native application observability

aspects. In the cloud-native paradigm, this changes: the life cycle of the network function service is separated from the hardware and software used to provide it – that is, what to achieve is separate from how to achieve it.

Cloudification separates shared infrastructure from shared platform services and dedicated (application) software. The separation of software from functional services has major consequences for both configuration and observability. Telemetry data is directly emitted independently from each disaggregated aspect of the system. There is no telecommunication node to provide aggregation. Instead, all telemetry data must carry its context as metadata to enable users to understand the system and get actionable insights.

To fulfill a service, infrastructure, platform and software resources must be orchestrated. In conformance with the cloud-native pet-and-cattle analogy, the continuation of the service (the what/the need) is what matters, while the hardware and software resources used to deliver it (the how) are disposable and interchangeable. As software instances take turns delivering the service, the software that is momentarily in use needs to emit telemetry both for the service it is currently delivering as well as for its own software observability. The platform and the infrastructure emit telemetry for their individual domains as well.

Because of the decoupling between the service and the software, telemetry for the service can be emitted by several different software instances over a period of time but still relate to the same service instance. Consequently, the service coupling needs to be provided by the emitted data itself; one cannot rely on the identity of any specific emitter to provide the service coupling. This behavior is not restricted to telecom – the same applies for software-as-a-service

MONITORING METHODS

The objective of the four golden signals [4] is to monitor distributed systems with user-facing metrics:

- Latency (the time it takes to service a request)
- Traffic (a measure of how much demand is being placed on the system)
- Errors (the rate of requests that fail)
- Saturation (how “full” the service is).

The objective of the USE method [6] is to identify systemic bottlenecks quickly in early performance investigations based on the provision of the following metrics by any resource:

- Utilization (percentage of time that the resource was busy)
- Saturation (amount of work the resource has to do)
- Errors (count of error events).

The objective of the RED method [5] is to monitor microservices based on the provision of the following request-centric metrics:

- Rate (the number of requests per second)
- Errors (the number of failed requests per second)
- Duration (the amount of time these requests take).

(SaaS) systems (or any aaS offer). The application instance execution is continuously served by software, but it should be possible to change the software instances that are used without noticeable impact on the application service delivery.



Target architecture for observability data management

Figure 3 provides an illustration of our high-level functional target architecture for observability data management, which is designed to overcome all of the challenges and implement the necessary changes to support cloud-native solutions. It is comprised of four key functional blocks and the interfaces between them.

Observability data consumption

The observability data consumption block at the top of Figure 3 is where machines and humans consume the data that has been exposed by the observability data ingestion, processing and exposure block. Any data consumption that requires data from more than one data source must access the data from the observability data ingestion, processing and exposure block. Operations support systems (OSS) functionality such as assurance and automation functions, as well as other tools used to observe the behavior and state of the application, are typical for this block, along with business support systems functionality and other analytics or observability tools. External users such as monitoring or post-processing systems from CSPs also fall into this category.

Observability data ingestion, processing and exposure

The role of this functional block is to collect data from different sources according to the “collect data once, use many times” principle. It stores and distributes data using the appropriate data services and executes data processing that can be application specific (using rules provided by the applications), generic (configurable filtering or aggregation, for example) or both. This block also exposes data over relevant legacy or cloud-native management interfaces and provides control capabilities to configure data collection, as

well as processing telemetry data from the realization layers and correlating/translating them into a functional view that is exposed through its northbound interfaces. Components of this functional block also fulfill the requirements of a data ingestion architecture [7].

The application

The application block is where business logic and supporting services run and execute their tasks, typically related to traffic handling, and where telemetry data generation takes place. Besides the actual business logic, there are typically telemetry agents that collect, filter and stream the data. Usually, there is also functionality that enriches the data records by adding metadata to enable traceability and to facilitate correlation on the receiving side. This block can be comprised of one-to-many workload component instances executing in one-to-many infrastructure clusters.

CaaS/infrastructure

The CaaS/infrastructure block serves as the execution environment for an application’s workloads. It primarily consists of K8s clusters, but it can also contain infrastructure-as-a-service (IaaS) and platform-as-a-service (PaaS) components as well as the hardware, depending on the deployment scenario. The multiple data sources of the platform and infrastructure telemetry data are located in this functional block and, as in the application, multiple telemetry agents collect and stream the data forward. This execution environment could be provided either in the form of data-center resources in a private cloud or as the corresponding services of a hyperscale cloud provider in a public cloud.

Interfaces

Several interfaces are needed to connect the functional blocks of the target architecture, namely:

Cloud-native application observability

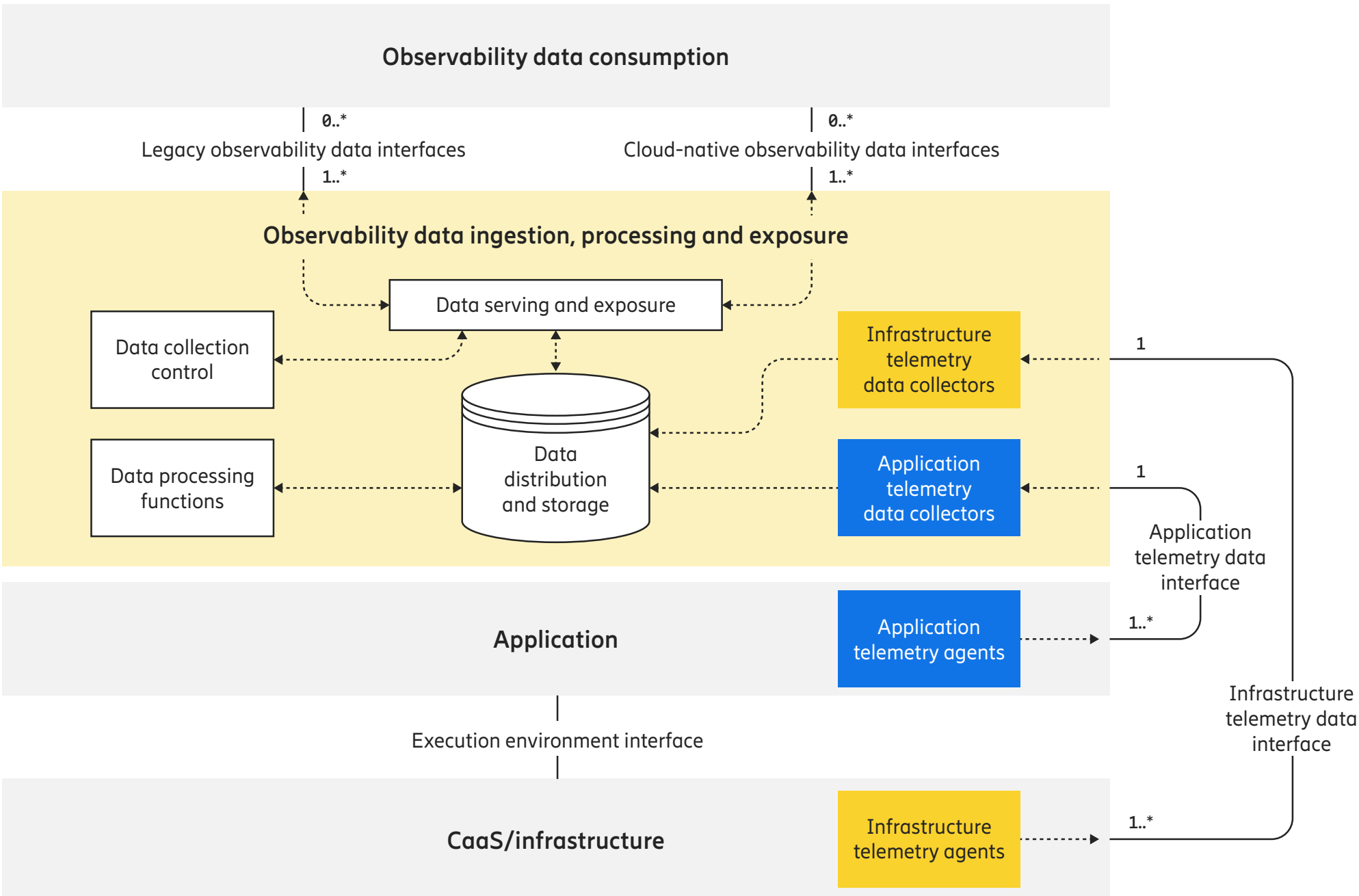


Figure 3: High-level functional target architecture

1. Legacy observability data interfaces
2. Cloud-native observability data interfaces
3. An application telemetry data interface
4. An infrastructure telemetry data interface
5. An execution environment interface.

Legacy observability data interfaces are needed to provide backward compatibility, which can be achieved with the help of different components that can expose services from different implementations compared with the

existing solutions. These interfaces can be standardized or proprietary and expose, for example, traditional performance measurement data files or fault management alarms.

Cloud-native observability data interfaces are new interfaces that enable a data management paradigm that supports cloud-native deployments. These interfaces include those that expose the observability data plane and those that expose the observability control plane. They can be standardized or proprietary. Typical data plane interfaces in

this group expose telemetry log, metric and trace data, and support queries on them all. Control plane interfaces provide management capabilities for observability data schemas or configuration capabilities for the data collection. The application telemetry data interface provides telemetry data (primary metrics/logs/traces/events) specific to the business logic.

The infrastructure telemetry data interface exposes telemetry data from the relevant infrastructure to correlate with the application to understand the functional behavior of the application business logic as well as to troubleshoot and determine the root cause for any deviation in functionality.

The execution environment interface is used by the application to retrieve the execution environment resource identifiers of the executing workloads. This is typically done through the K8s application programming interfaces (APIs). Note that these interfaces do not expose the infrastructure telemetry data, but rather the metadata needed to enrich the application telemetry data.

All of these interfaces require access control to ensure that only authorized users can access the specific data to which they are entitled. Furthermore, some of the interfaces may need to be visible only for certain user domains or tenants to enable a separation of public and non-public data exposure.

The OpenTelemetry project

The CNCF hosts a variety of open-source projects that are widely used in the industry and have made significant contributions to the development and standardization of cloud-native technologies, not least within the observability and analysis area. Ericsson has adopted several of these projects over the years.

The CNCF successfully incubated and graduated the Prometheus, Fluentd and Jaeger projects, achieving a high level of vendor neutrality from telemetry data instrumentation to telemetry data collection. However, those projects were essentially implemented as verticals with their own software development kits (SDKs), telemetry collectors and limited data alignment, which resulted in significant footprint and usability challenges. There were also open-source projects such as OpenCensus and OpenTracing, which had similar objectives but did not work well together, resulting in duplicated and wasted efforts.

The OTEL project is relevant to the telecom domain for several reasons.

The OpenTelemetry (OTel) project was started with the objective of delivering a horizontally-aligned telemetry framework across the three observability pillars, while keeping the initial vendor-neutral objective. OpenCensus and OpenTracing were sunsetted and merged into OTel, which is now actively driven by the CNCF.

Figure 4 illustrates the scope of the OTel project, which has three key aspects:

- 1. It secures the availability of a unified telemetry instrumentation API and SDKs in all common programming languages.
- 2. It provides common data models and semantic conventions.

Cloud-native application observability

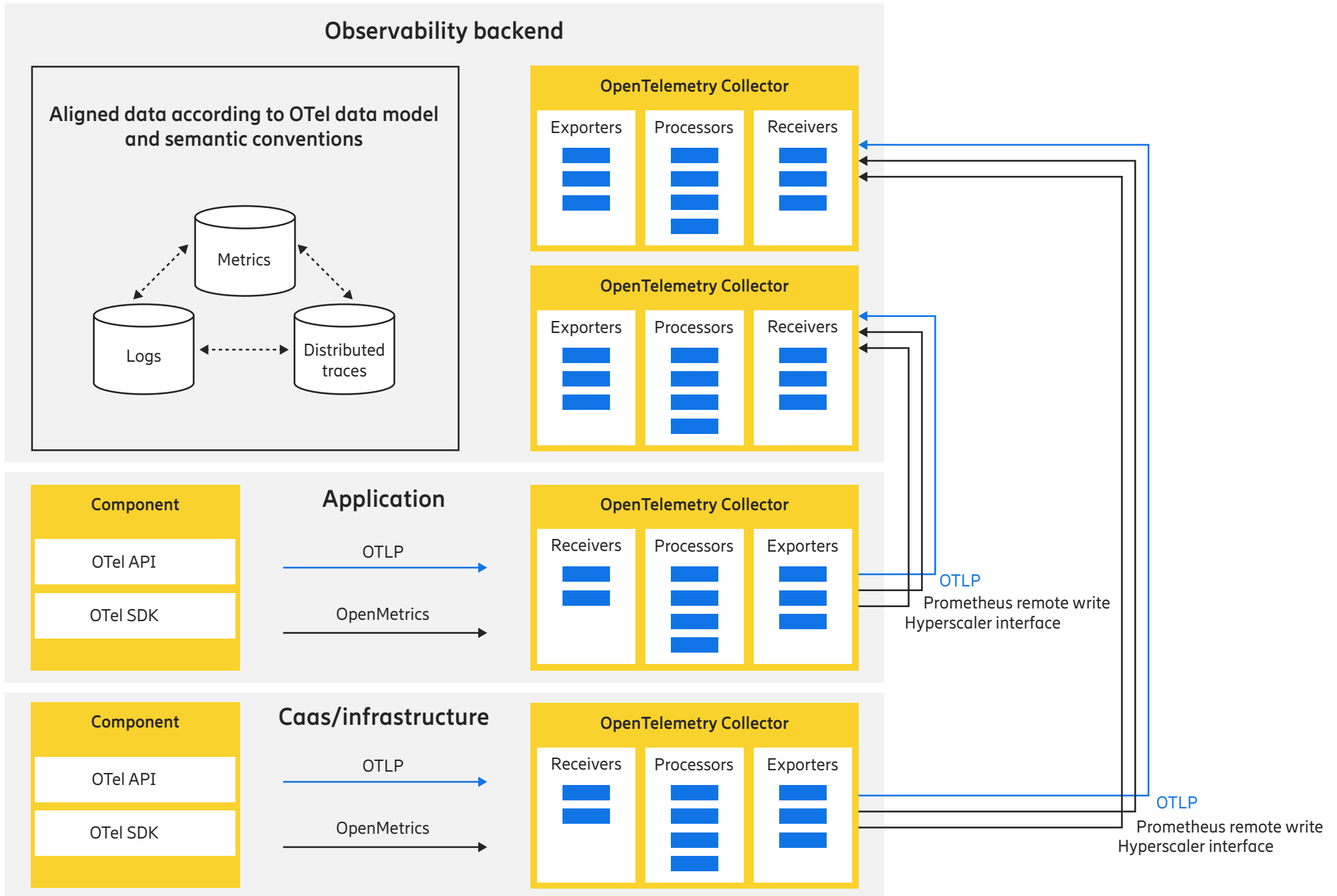


Figure 4: The scope of the OTel project

- 3. It provides efficient ways to collect, process and forward telemetry data thanks to the new OpenTelemetry Protocol (OTLP) and the very modular OTel Collector component.

The OTel project is relevant to the telecom domain for several reasons. Firstly, the OTel Collector stateless agent deployment model, with a clear separation from stateful backend, will enable application footprint savings. Secondly, the ability to pick/choose/design suitable OTel Collector

processors will help to reduce the emitted data volume. Thirdly, the end-to-end metric data push will result in latency reduction for metrics-based closed-loop automation. Lastly, data alignment will lead to a reduction in mean time to resolve.

Key benefits of cloud-native application observability

Cloud-native application observability can both improve existing use cases and enable new ones by:



1. Enhancing operational efficiency
2. Reducing integration effort
3. Enabling observability of distributed and disaggregated systems.

Enhancing operational efficiency

Cloud-native application observability enables significantly faster access to critical data. In contrast to legacy exposure mechanisms employed in telecoms, such as periodic file collection, cloud-native application observability operates at near-real-time speed. This reduction in latency of data collection in turn reduces the time to detect and diagnose suboptimal or undesirable network conditions, thereby facilitating quicker adjustments to the network.

Cloud-native application observability enables significantly faster access to critical data.

Faster data access is an essential prerequisite to effective automation, as it exposes a view of the network that is temporally proximate to the current state of the network. Use cases that can benefit include decisions around autonomous scale-in/scale-out, where quick decisions about the former can reduce the risk of wasted resources, while timely decisions about the latter can help to ensure that required capacity is available when needed. Decisions about power utilization and optimization can similarly benefit from timely access to data.

The opportunities presented in terms of operational benefits do not come without trade-offs and compromises. There is a capital expense in terms of compute, storage and networking resources to generate, transport and process telemetry data. Low-latency, streamed data may not be buffered/persisted for replay at source, which means that robust data management is essential to minimize the risk of data loss that would impede effective decision-making. Judicious usage of the telemetry control plane to manage data is required to deliver an acceptable balance between sufficient observability and manageable data volumes.

Reducing integration effort

The exposure mechanisms used by telecom applications at present comprise many diverse data encoding approaches. Observability data is exposed using a mix of encoding and transport mechanisms with SNMP (Simple Network Management Protocol) typically used for alarm/alert-style data, file or TCP (Transmission Control Protocol) streams used for metrics and a disparate set of (non) standards for log and trace data. Telecom equipment vendors and CSPs incur recurring costs to develop, support and operate networks that expose data in this fashion.

Cloud-native communities, such as the CNCF, actively encourage the use of standardized approaches to observability data management. Common aspects of metrics, logs and distributed traces are handled in a common fashion; the schemas used to encode/decode the data are common, as is the transport mechanism. This reduces the effort to embed the capability for telecom equipment vendors within their offerings and enables easier integration into tool chains that are designed to utilize such data. OpenTelemetry is fast becoming the de facto standard in this space.

Cloud-native application observability

Enabling observability of distributed and disaggregated systems

As noted, cloud-native applications may be deployed in a distributed manner within a certain context (such as a namespace). The use of OTel APIs, SDKs and tools, which inculcate the use of effective distributed tracing, eases the correlation of data across applications deployed using a distributed architecture.

Similarly, the use of cloud-native application observability facilitates the correlation of application telemetry with telemetry sourced from the typically independently provisioned and operated runtime infrastructure on which the applications are deployed. Frameworks such as OTel and cloud-native observability techniques are being adopted by cloud-native infrastructure providers, including hyperscalers. Embracing an aligned approach across the application and infrastructure layers reduces the challenge of managing and correlating data across application and infrastructure, thereby reducing R&D spending.

Conclusion

Ericsson agrees with the cloud-native community view that the purpose of application observability is to provide actionable insights into an application’s performance and characteristics. These insights, gained by correlating data from different, independent and self-contained data sources, can be used in assurance loops that optimize the application’s behavior, do trend analysis to predict future needs and, in case of some anomaly, troubleshoot the application. We see this approach as crucial to achieve our goal of a highly automated network managed by closed-loop telecom applications driven by artificial intelligence.

We acknowledge the disaggregated nature of the telecom applications and the consequences with respect to the continuity of the application being captured in its data as opposed to the continuity of its individual parts. Therefore, the collection, persistence, processing and exposure of the telemetry data needs to be changed, utilizing cloud-native technologies such as those available as de facto standards from the open-source ecosystem. Aligned data exposure better supports correlation, post-processing and consumption of the data originating from multiple, disaggregated sources, such as applications, platforms and infrastructure.

Ericsson has a strong history of providing effective observability for our applications. To protect and grow that capability into the future, we must evolve our observability solutions together with our customers, by embracing these new and exciting cloud-native paradigms and technologies.

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Hybrid 5G Core network architecture with full 4G and 5G device support

Authors:
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In the future, when most devices are served by the 5G Core network, communication service providers will need an efficient way to manage legacy devices. Ericsson’s approach simplifies the network architecture and harmonizes the service offering by creating a hybrid core network that can support legacy 4G and 5G non-standalone devices alongside 5G standalone devices.

Hybrid 5G Core networks



Mobile network architecture evolution in the 2020s has been characterized by the initial 5G buildout and the ongoing transition to the 5G System (5GS) based on 5G New Radio (NR) standalone (SA) architecture and the 5G Core (5GC) network.

By 2028, it is projected that the global population coverage of 5G will reach 85 percent and the majority of traffic will be handled by 5G networks [1]. Network support for 2G and 3G will soon be phased out, as those communication service providers (CSPs) that have not already done so are currently in the process of shutting these networks down.

To accelerate the adoption of 5G services, new 5GC-enabled capabilities and Service-Based Interfaces (SBIs) will be needed. The SBIs will replace some of the legacy Diameter interfaces and support new charging models defined in the 5GS. This migration presents an opportunity to harmonize policy and charging across 5G non-standalone (NSA) and 5G SA devices and thereby reduce operational cost.

CSPs will also need the capability to handle legacy 4G devices efficiently from both a core network architecture

perspective and an operational perspective all the way to the end of the 4G era when most spectrum will be allocated to 5G and capacity shifts to handle 5G – and eventually 6G – traffic. Examples of 4G legacy devices that will require support include low-cost feature phones and long-lived Massive Internet of Things (M-IoT) devices such as the low-power wide-area (LPWA) devices used in utility companies’ power meters and in cars with 4G access. Many of these devices have lifetimes that are measured in decades. The most efficient way to prepare the core network architecture for the eventual sunset of 4G is to use the network functionality provided by the 5GC.

The hybrid core architecture we propose is designed to provide future-proof support for the long tail of legacy 4G and 5G NSA devices in the network. Based on the 5GC, with common subscription management, policy and charging, our hybrid core architecture proposal minimizes the need for system integration, optimizes operations and aligns with 5GC ways of working, including the use of SBIs. It is even possible to decommission some legacy Evolved Packet Core (EPC) nodes with no impact on existing 4G radio access networks (RANs) and devices.

Terms and abbreviations

3GPP – 3rd Generation Partnership Project | **5GC** – 5G Core | **5GS** – 5G System | **AMF** – Access and Mobility Management Function | **APN** – Access Point Name | **AS** – Access Stratum | **CCS** – Converged Charging System | **CHF** – Charging Function | **CSP** – Communications Service Provider | **DNS** – Domain Name System | **EPC** – Evolved Packet Core | **E-UTRAN** – Evolved Universal Terrestrial Radio Access Network | **FWA** – Fixed Wireless Access | **gNB** – gNodeB | **HSS** – Home Subscriber Server | **IMS** – IP Multimedia Subsystem | **LPWA** – Low Power Wide Area | **LTE** – Long Term Evolution | **M-IoT** – Massive Internet of Things | **MME** – Mobility Management Entity | **NAS** – Non-Access Stratum | **ng-eNB** – Next-Generation eNodeB | **NG-RAN** – Next-Generation Radio Access Network | **NNI** – Network-to-Network Interface | **NR** – New Radio | **NSA** – Non-Standalone | **OCS** – Online Charging System | **OFCS** – Offline Charging System | **PCF** – Policy Control Function | **PCRF** – Policy and Charging Rules Function | **PDN** – Packet Data Network | **PDU** – Protocol Data Unit | **PGW** – Packet Data Network Gateway | **PGW-C** – PGW Control Plane Function | **PGW-U** – PGW User Plane Function | **RAN** – Radio Access Network | **SA** – Standalone | **SBI** – Service-Based Interface | **SGW** – Serving Gateway | **SMF** – Session Management Function | **UDICOM** – User Data Interworking, Coexistence and Migration | **UDM** – Unified Data Management | **UPF** – User Plane Function

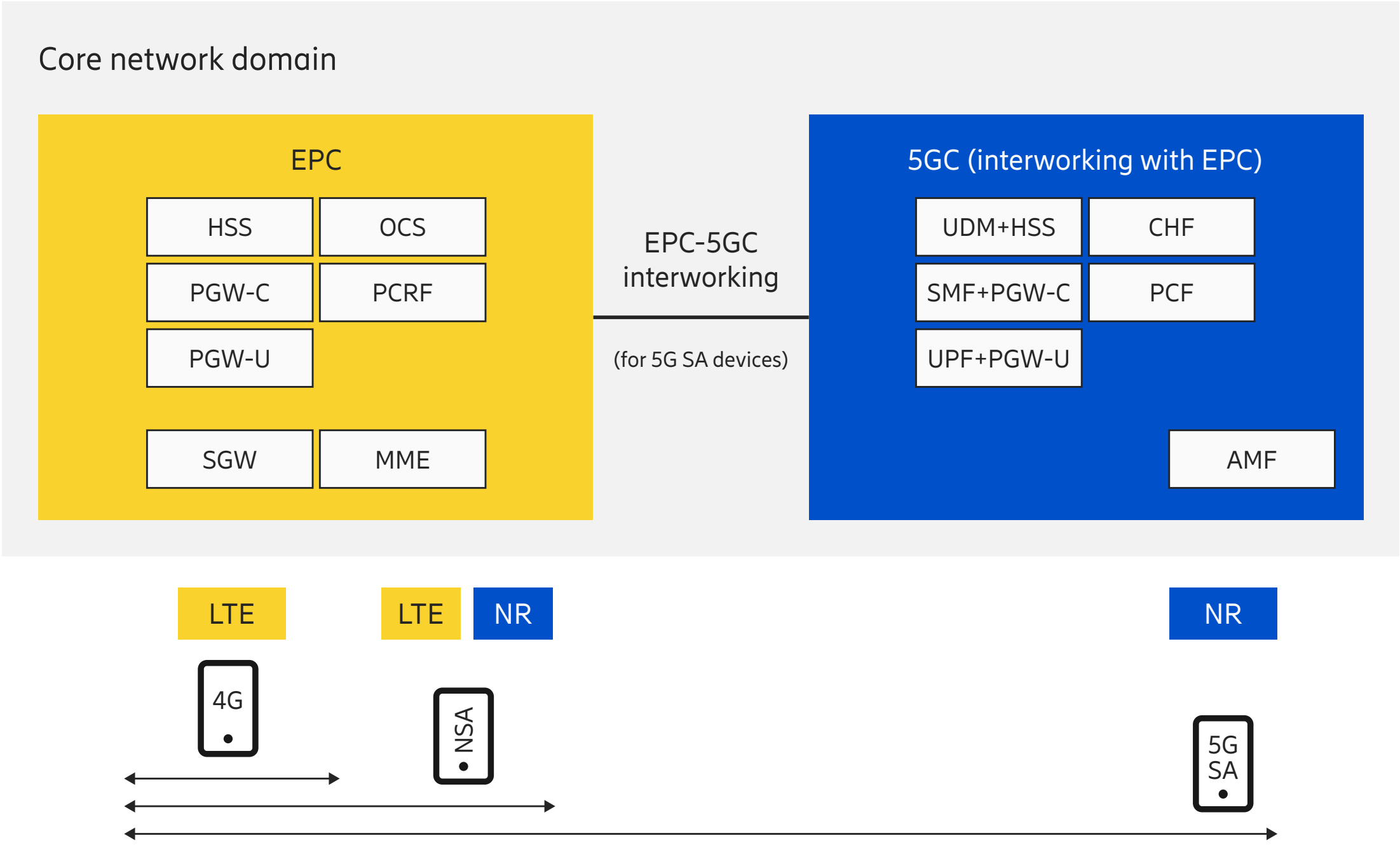


Figure 1: EPC-5GC interworking architecture

5GC-capable devices versus legacy 4G devices

The 5GS was initially specified for devices that have the corresponding support – that is, devices that support all required functionalities and in particular interfaces to interact with the next-generation RAN (NG-RAN) and the 5GC. These devices interact directly with the gNodeB (gNB) in the NG-RAN and with the Access and Mobility Management Function (AMF) in the 5GC. The communication layer for wireless communication is between

the device and the gNB in the NG-RAN known as the access stratum (AS), which includes the Radio Resource Control.

The communication layer between the device and the AMF is called the non-access stratum (NAS). All signaling related to registration in the 5GC, protocol data unit (PDU) session establishment and configuration updates from the AMF are communicated through NAS. We refer to devices that support this type of communication with the AMF as “5GC-

Hybrid 5G Core networks

capable devices.” The terms “NR SA devices” and “5G SA devices” are used synonymously.

Legacy 4G devices use EPC NAS to communicate with the Mobility Management Entity (MME) and thereby to connect to EPC over LTE (Long Term Evolution). 5G NSA devices are also considered to be legacy 4G devices from a core-network perspective because they use EPC NAS and not 5GC NAS.

The EPC-5G Core interworking architecture

To enable a smooth introduction of the 5GS, the 3GPP (3rd Generation Partnership Project) has defined an interworking architecture [2] between the EPC and 5GC to support 5GC-capable devices that need to move to 4G in cases where the 5G SA coverage is not fully built out, as shown in **Figure 1**.

The interworking architecture enables the device to retain the IP address during mobility between the Evolved Universal Terrestrial Radio Access Network (E-UTRAN) (4G/5G NSA) and NG-RAN (5G SA) as a prerequisite for uninterrupted service, in particular for, but not limited to, voice. The connections for data transmission (PDU sessions if established through the NG-RAN and a packet data network (PDN) connection if established via the E-UTRAN) are always anchored in the SMF+PGW-C (session management function + packet data network gateway control plane) of the 5GC. The 5GC-capable devices – marked as 5G SA devices in Figure 1 – are always anchored in the 5GC, regardless of whether they are connecting over 4G or 5G access. Legacy 4G and NSA devices are anchored in the EPC, which includes using the Home Subscriber Server (HSS).

When a 5GC-capable device is accessing the network over E-UTRAN, the MME is responsible for selecting an SMF+PGW-C as the session anchor point. This occurs during

PDN connection establishment if the HSS subscription data allows the session to be anchored in the 5GC and the device supports communication with the 5GC. When all of these criteria are fulfilled, the MME performs a special query toward the Domain Name System (DNS) to find the address of an SMF+PGW-C. In the message to create the session, an indication is set for the 5GC, which thereby dictates the use of the 5GC SBIs – in this case, N7 and N40 for policy and charging respectively.

Extending the anchoring possibilities to legacy 4G and 5G NSA devices

In Release 16, the 3GPP enhanced the EPC-5GC interworking architecture by extending the anchoring possibilities beyond 5GC-capable devices to also include legacy 4G devices (not supporting 5GC NAS) in a 5GC gateway (SMF+PGW-C). This improves the network evolution flexibility. We call this the hybrid core architecture since it supports the migration of legacy 4G and 5G NSA device types, enabling them to connect to the 5GC.

The new capability for the SMF+PGW-C is to support parameter conversion between the EPC and 5GC through the assignment of a device-unique PDU Session ID per PDN connection on behalf of the device. This PDU Session ID is used internally inside the 5GC network – for example, toward Unified Data Management (UDM), but the PDU Session ID is not sent to the legacy 4G and 5G NSA devices (these devices would be unable to use it).

The MME is enhanced to facilitate granular selection of an SMF+PGW-C for legacy 4G and 5G NSA devices in case legacy PGWs are still present in the network. If not, there is no need for the MME to differentiate. The selection logic may also support forcing a device on a specific Access Point

Name (APN) to SMF+PGW-C, while using a legacy PGW for other APNs. This enables a stepwise migration to 5GC per APN. The resulting network architecture is presented in **Figure 2**, which shows the main core network functions and interfaces supporting the 4G and 5G NSA devices.

Assuming that the network is configured correctly and that the subscription data for the user exists in the UDM of the 5GC, implementation of the hybrid core architecture will result in the following sequence for PDN connection establishment.

The hybrid core architecture supports the migration of legacy 4G and 5G NSA devices to connect to the 5GC.

The procedure to establish a PDN connection is initiated by the device through the RAN to the core network in the same way a 5GC-capable device would connect over LTE. During this procedure an IP address is assigned to the device for that specific connection. Once the procedure is successfully concluded, the device can send and receive data from external sources. For the core network to correctly establish the connection, the device will provide an APN to the core network. This APN will be crosschecked toward the subscription data in the HSS, assuring that the user is authorized to use the service.

The MME then uses the APN in an extended query sent to the DNS together with an indicator that a hybrid core solution is wanted (an enhancement of the MME). The DNS responds with the address to the session anchor point, SMF+PGW-C. The MME uses the geographical location of the device to find a serving gateway (SGW).

A message is then sent from the MME through the SGW to the SMF+PGW-C, to allocate resources and establish a connection for this specific device. In this message, the MME informs the SMF+PGW-C, that anchoring the connection in the SMF+PGW-C is allowed by the subscription data.

The SMF+PGW-C selects a UPF+PGW-U instance, and assigns an IP address and a PDU Session ID. The latter is used by the UDM, policy control function (PCF) and charging function (CHF) to identify the user and the session. The SMF+PGW-C uses a network repository function (NRF), which is a fundamental part of the Service-Based Architecture, to discover the UDM, PCF and CHF. Once all resources have been successfully allocated by the core network, the assigned IP address will be returned to the device in the response to the initial request to establish a PDN connection.

Impact on adjacent architecture domains
The hybrid core architecture includes changes to multiple architecture domains, allowing the management of legacy 4G and 5G NSA devices in the same way as 5G SA devices. As shown in Figure 2, the 5GC SBI N7, N10, N28 and N40 are used to manage user profiles, policy and charging.

Subscription and policy information in the 5GC user data repository must also allow legacy 4G and 5G NSA devices to use the UDM and PCF. This solution makes it possible

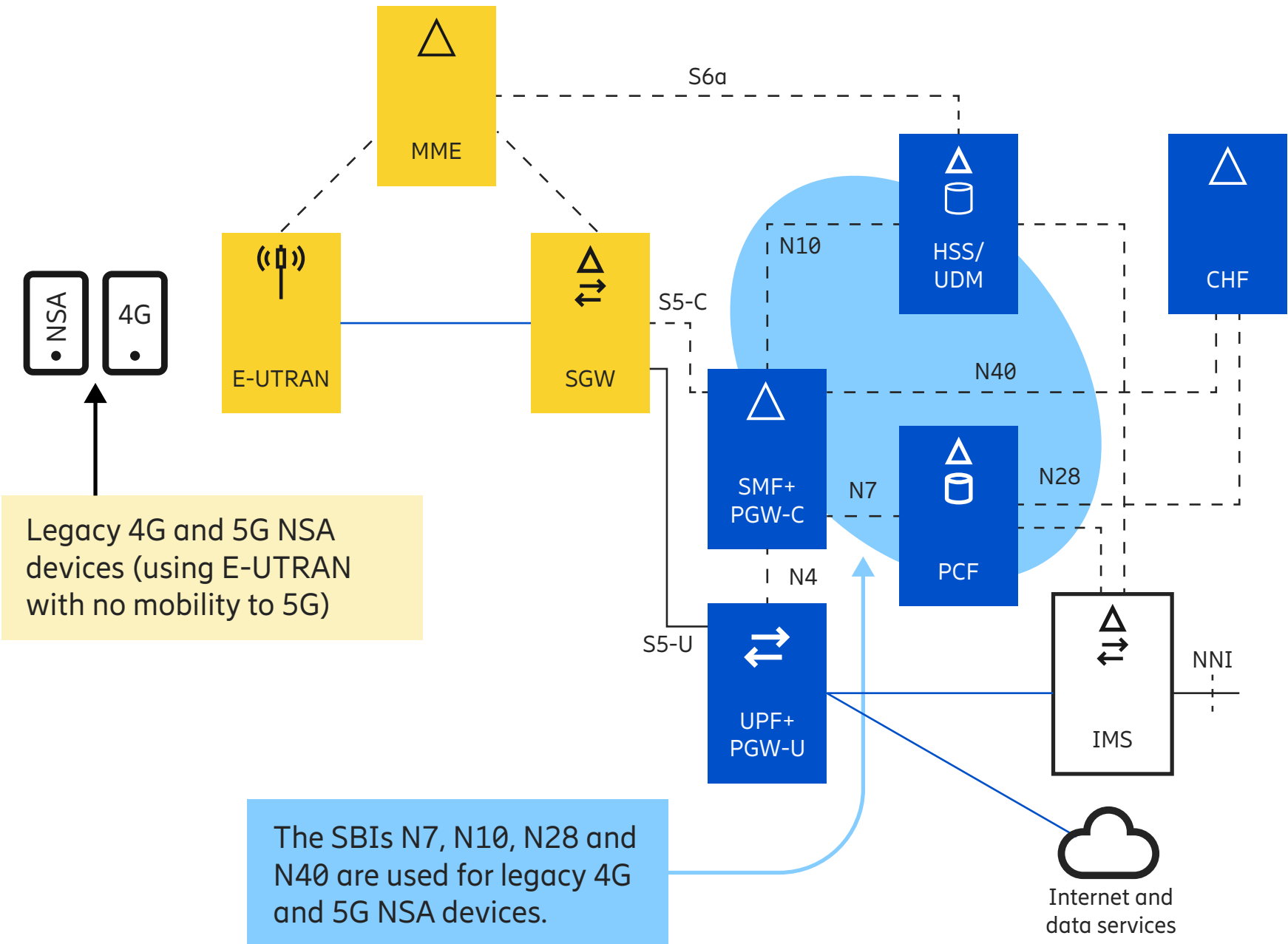


Figure 2: Key concept of the hybrid core architecture

to manage both new 5GC-enabled subscribers regardless of the access technology used and legacy 4G users who connect only through the E-UTRAN in a common solution. The subscription data update includes information for the MME to select the SMF+PGW-C in the 5GC as the anchor point. This information is available to the MME through the HSS part of the solution (over the legacy Diameter interface S6a).

Deployment of the 5GC implies that the online charging system (OCS) and offline charging system (OFCS) are modernized to a 5GC converged charging system (CCS). This makes it possible to remove several legacy Diameter interfaces and replace them with SBI-based 5GC interfaces such as N28 (PCF to CHF) and N40 (SMF+PGW-C to CHF). Adding legacy 4G and 5G NSA devices to the CCS implies that the CHF part of the CCS is responsible for all charging data records generation in the 5GC, including all traffic



for the legacy 4G and 5G NSA devices. Online rating and balance management of the 4G and 5G NSA devices’ traffic works the same way as it did in the OCS.

Additional architecture considerations

The hybrid core architecture is the first step toward fully embracing the new 5GC architecture. However, the existing EPC network functions PGW and PCRF (policy and charging rules function) as well as OCS/OFCS may still be needed for purposes such as outbound roaming, supporting other business verticals and/or M-IoT that depend on 2G/3G radio network usage. For parts of the legacy network, it will also be necessary to keep existing Diameter interfaces. For example, Diameter is used for IMS (IP Multimedia Subsystem) as well as in the packet core between an MME and HSS (S6a) for attach and mobility procedures.

One aspect of the 3GPP-specified EPC-5GC interworking architecture that is also applicable to the hybrid core is that there is no IP session continuity if mobility from 4G to 2G/3G occurs, which depends on the 4G coverage. The signaling load could increase if this mobility happens too frequently. If the CSP has shut down 2G/3G or has full coverage in suitable bands for 4G, this will not be a problem.

Key benefits

The hybrid core architecture delivers significant value compared with the device connectivity option of using 5GC NAS over LTE that was specified by the 3GPP but not adopted in the industry (known as “option 5 with ng-eNB”). That option would require a major network upgrade including new devices and LTE RAN, and even then it would not support legacy 4G devices as they are not 5GC NAS capable.

A key benefit of the proposed hybrid core architecture is that it makes it possible to modernize user profiles, policy and charging by using single policy and charging functionalities in the 5GC for legacy 4G devices, 5G NSA devices and 5GC-capable devices. This provides opportunities for savings in system integration and in total cost of ownership, enabling continued network and service evolution.

Another benefit of the proposed architecture is that the migration of each use case to the hybrid core offloads the legacy nodes. Once all use cases have been migrated to the hybrid core and any additional architecture considerations described above have been taken into account, network deployment can be simplified by decommissioning the legacy PGW, charging and policy functionality. As a result, the associated Diameter-based interfaces such as Gx (policy control), Gy (online charging) and Sy (spending limit reporting) will no longer be needed.

The migration of 5G NSA devices to the hybrid core architecture could be regarded as a step toward 5G SA, as the devices will be transferred to a common subscription management, policy and charging solution based on the 5GC. This step will facilitate a transition from Diameter to SBI-based charging and spending limit control using the 5GC charging architecture. That will help familiarize CSPs with the operational methods of SBI-based charging architecture and prepare for the addition of new charging trigger points that the 5GC architecture brings, enabling more advanced charging schemes. Examples of such trigger points are related to service exposure, network slice admission control, network slice management and data analytics.

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Once a CSP has successfully launched 5G SA to handle the majority of its own mobile broadband subscribers, or perhaps even started the transition to 6G, there will be a long tail of 4G devices that still require support. These can be the CSP’s own subscribers with feature phones, inbound roaming devices or 4G M-IoT devices (NB-IoT and Cat-M). The M-IoT business is characterized by long usage cycles, where CSPs are offering contracts to automotive companies, utilities and others with up to 20 years of device support. Another key aspect to consider is the need for low-cost radio modems in M-IoT devices, driving the requirement on simple architecture and high volumes. The hybrid core network architecture is a cost-efficient solution for these types of devices.

The migration process

A CSP that wants to implement the hybrid core architecture must start by taking a use-case perspective and deciding which device categories and APNs to move for that particular use case. Second, it must consider a list of prerequisites for the selected use case prior to migration. Third, it must perform the actual migration for the device categories and APNs. Finally, it needs to perform post-migration actions.

A CSP typically supports a large variety of use cases for its own subscribers on 4G, including consumers, enterprises, automotive and fixed wireless access (FWA), for example. A CSP may decide to migrate all APNs applicable for a use case to the hybrid core, or it may decide to migrate only one APN – enterprise, for example – to hybrid core, leaving other APNs on legacy PGWs.

If a CSP decides to support legacy 4G devices on the hybrid core, for example, it may choose to migrate one use

case after the other. Prior to each migration, it needs to determine the prerequisites for that particular use case and then perform the actual migration. For example, a CSP may choose to migrate FWA devices first, then enterprise devices, followed by consumer smartphone devices. Once the CSP has migrated the first use cases, it may not be necessary to perform every step again.

A key benefit of the proposed hybrid core architecture is that it makes it possible to modernize user profiles, policy and charging.

Our checklist of prerequisites for the migration is based on the architecture depicted in Figure 2. One of the first steps is to provision the subscription data for the legacy 4G and 5G NSA devices in the 5GC UDM. For these devices there is subscription data accessed through the HSS and the UDM. The legacy HSS is typically evolved to support the UDICOM (NU1) interface to the UDM. As a result, the legacy HSS deployment realizes the HSS part in the HSS+UDM function shown in Figure 1 and is no longer needed as a separate network function.

The required capabilities to handle the particular use case must be enabled in the SMF+PGW-C, the MME must be enhanced and the DNS configured to select an SMF+PGW-C for legacy 4G and 5G NSA devices.

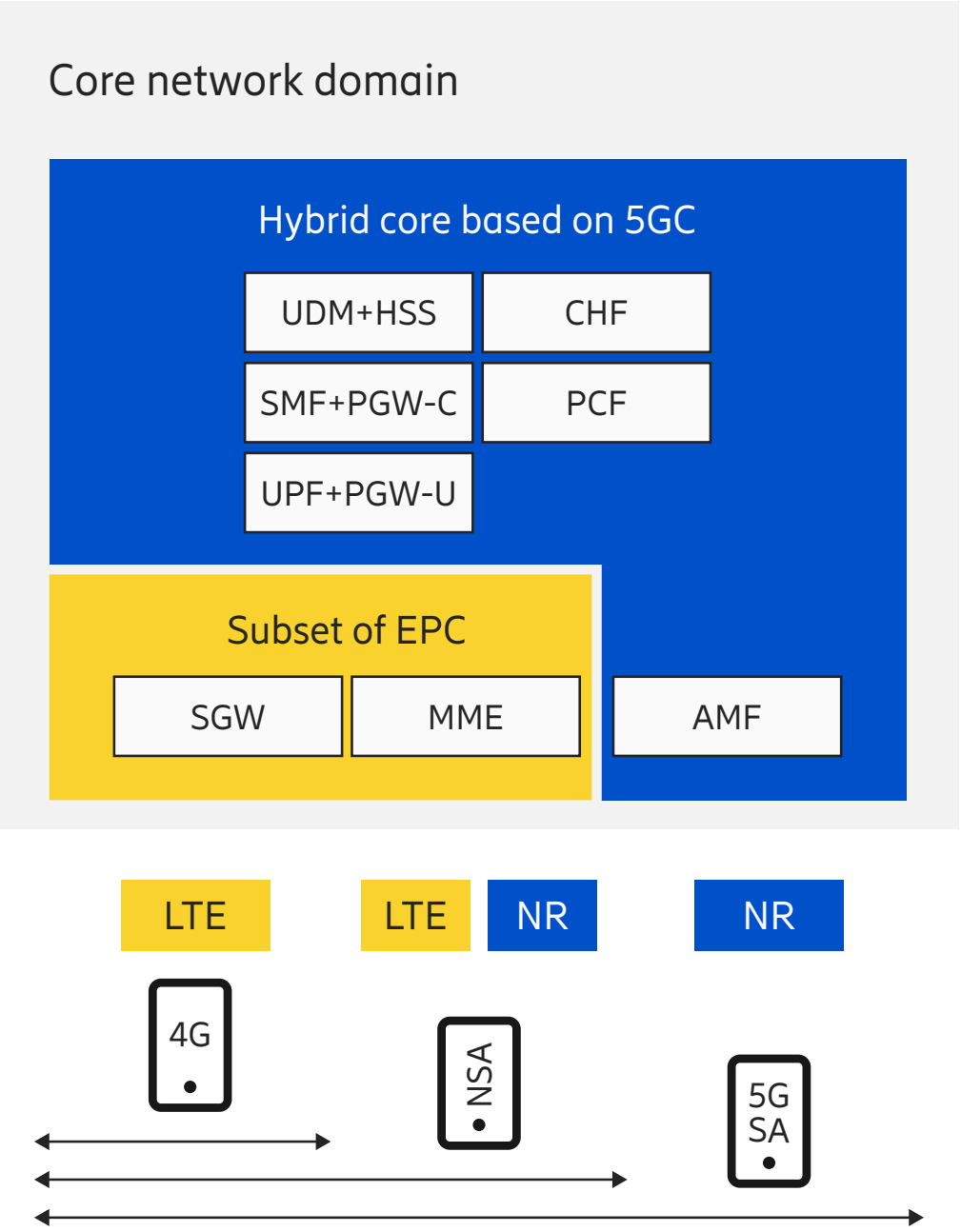


Figure 3: Network simplification through migration to a hybrid core architecture

The capacity needs on legacy nodes may reduce before the migration is finished. However, the CSP may only start to decommission legacy nodes that are no longer needed (such as PGW-C, PGW-U, OCS and PCRF) once all use cases (and therefore all devices) have been migrated to the hybrid core. When the decommissioning is complete, the network simplification shown in **Figure 3** will be the result. When this simplification has been achieved, the core network domain includes only the minimal set of network functions needed

to serve all the devices shown in the figure. In particular, only the SGW and MME are mandatory as a subset of EPC to maintain the interface toward E-UTRAN and 4G devices.

Conclusion

Migration toward our proposed hybrid core architecture offers many benefits to communication service providers (CSPs) that want to modernize their network architecture without introducing disruptive ecosystem changes late in the

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4G life cycle. We believe that architecture evolution should not require any changes to the legacy 4G and 5G non-standalone (NSA) devices or to the already widely deployed LTE (Long Term Evolution) radio access network.

Based on features specified in 3GPP (3rd Generation Partnership Project) Release 16, our hybrid core architecture proposal makes it possible to anchor legacy 4G and 5G NSA devices on the 5G Core (5GC), creating the opportunity to apply 5GC operational methods for handling subscription, policy and charging to both 5GC-capable and legacy 4G and 5G NSA devices. It further provides possibilities for savings in system integration and operational costs when aligning with the 5GC ways of working, such as using Service-Based Interfaces instead of Diameter interfaces.

The migration of 5G NSA devices to the hybrid core architecture is the most obvious early use case, as the adoption of 5GC capabilities delivers a variety of benefits that enable continued network and service evolution. In the longer term, CSPs will want to use the hybrid core architecture to secure cost-efficient support for the long tail of legacy 4G devices in the network, including Massive Internet of Things devices. Hence, the hybrid core architecture provides a single solution for all 4G and 5G devices.

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Hybrid 5G Core networks



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Beyond bit-pipes — new opportunities on the 6G platform

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The 6G platform will enable communication service providers to take a major step toward delivering more than best-effort communication. By enabling fast and easy introduction of new exposed services, 6G will be a powerful platform for developers to realize advanced use cases such as immersive communication.

6G platform opportunities



In the digital context, a platform is a product that enables other products or services to be built on top of it. Platform businesses typically enable others to share information, build applications or content, or run services tailored specifically for them. The 6G platform aims to expand the scope of networks beyond communication services to enable the delivery of broad support for applications, which are at the center of the digital transformation.

Enhanced access to network capabilities through service application programming interfaces (APIs) available in the 6G platform will make it easier to meet the high demands of applications on communications and other capabilities from underlying infrastructure. For communication service providers (CSPs), this will create opportunities to offer new services and create relationships with new customers that can be monetized based on service differentiation. With the 6G platform the opportunities for new monetization established in 5G will be further enhanced.

The mobile communication ecosystem today

Mobile communication exists in a wide and flourishing ecosystem that consists of several stakeholders who fulfill different roles. Two of the most important ones are CSPs and application service providers (ASPs). ASPs deliver services to private or enterprise consumers through software functionality (applications) in devices and in the internet

domain. Applications are developed and made available to consumers on application platforms (Google Play, for example) that serve as marketplaces, also allowing third-party enterprises to reach consumers with advertisements and other types of content. CSPs deliver a best-effort mobile broadband (MBB) data connection (also known as a bit-pipe) between consumer devices and the internet domain. The current business model for CSPs is based on consumer subscriptions, with applications relying on generic MBB.

Looking ahead, both current and future applications will require enhanced communication services and services beyond communication that enable application differentiation and the ability to improve user experience. For example, information services can help applications optimize for the user's position and current network conditions. Quality of service (QoS) can significantly improve user experience, helping the application move from best effort toward service differentiation.

While the traditional ASP and CSP roles are likely to persist into the 2030s, relationships between them will need to evolve to overcome emerging challenges. For example, both ASPs and CSPs already face limitations to their ability to innovate: ASPs because they are forced to adjust their services to an MBB bit-pipe and CSPs because of a lack of established business with the ASPs. Further, ASPs that want to use more than an MBB bit-pipe need to manage interactions with many CSPs.

Terms and abbreviations

3GPP — 3rd Generation Partnership Project | **AI** — Artificial Intelligence | **API** — Application Programming Interface | **AR** — Augmented Reality | **ASP** — Application Service Provider | **CSP** — Communication Service Provider | **GNSS** — Global Navigation Satellite System | **HMD** — Head-Mounted Display | **MBB** — Mobile Broadband | **MLOps** — Machine Learning Operations | **MR** — Mixed Reality | **NEF** — Network Exposure Function | **NW** — Network | **QoE** — Quality of Experience | **QoS** — Quality of Service | **SCEF** — Service Capability Exposure Function | **SDK** — Software Development Kit | **SLA** — Service Level Agreement | **URSP** — UE Route Selection Policy

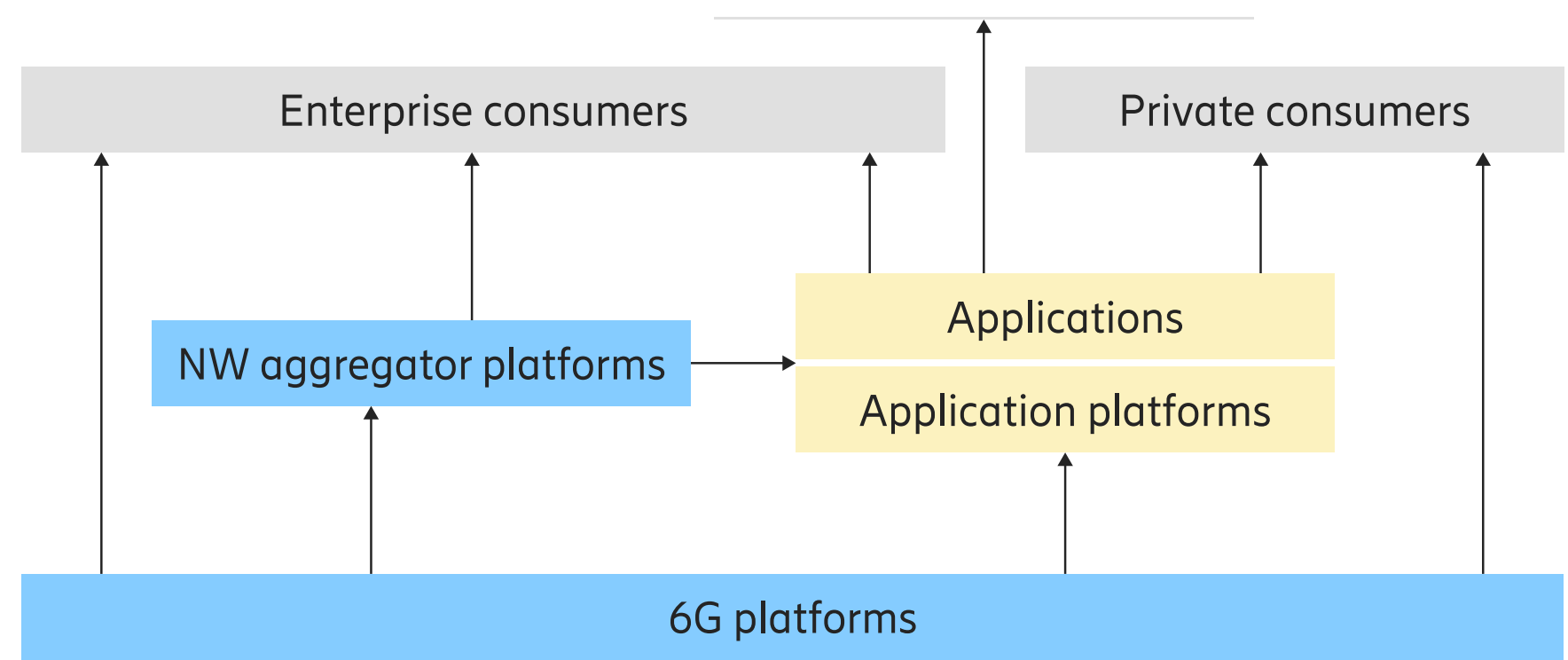


Figure 1: Interactions on the 6G platform

The best way to overcome these challenges is for CSPs to offer differentiable add-on services both directly to ASPs as well as through aggregator and application platforms, in addition to the MBB bit-pipe they offer to enterprise and private consumers. **Figure 1** shows how the 6G platform will incorporate service relationships with new stakeholders (applications and application platforms, in light yellow) next to those that already exist in 5G (with network aggregator platforms and directly with enterprise and private consumers). The offering for ASPs must be adapted to their roles in the ecosystem and be based on a solid understanding of their specific needs. To prevent fragmentation, it will be necessary to have a developer-friendly global interface that is coupled with attractive business agreements.

New service offerings on the 6G platform

The 2030s will be a decade of cyber-physical interactions – that is, the connections between the physical domain of things and actions and the digital domain of data and intelligence will intensify. Networks will serve as the key link between these domains, and we expect to see many new use cases that exploit these interactions.

Figure 2 illustrates a network platform built on 6G technology and services. Based on the network assets – that is, a set of available enabler technologies – this platform has capabilities of service assurance and enhanced external interactions for ASPs and enterprises. Such a platform enables a variety of new business possibilities.

6G platform opportunities

The top of figure 2 highlights a selection of the foreseen 6G platform use cases, all of which require improved performance, advanced services, efficiency and sustainability, based on new services offered to consumers through ASPs. Many of these use cases will benefit from the 6G network’s ability to offer capabilities beyond communication such as positioning, physical environment sensing and compute. These capabilities reuse existing communication assets for add-on services to data connectivity.

Mixed reality (MR) [1] is a good example of a demanding use case. In MR, digital content is seamlessly integrated into a user’s view, bringing to life distant places and making it possible to interact with others as if they were in the same physical space. This use case requires high-quality image rendering with low latency and precise spatial mapping of digital objects. At the same time, the user device should be lightweight and affordable, with good battery performance. To achieve this, the 6G platform can support devices and ASPs at multiple levels: communication with quality guarantees, localization and mapping support, and offload services for computational tasks and artificial intelligence (AI).

The role of network platforms

Communication is a cornerstone of all forms of digital business, with networking capabilities setting the boundaries for what can be achieved. Beyond serving as communication tools, today’s networks can also provide:

- Tailored and engineered connectivity services, such as QoS guarantees
- Information services from traditional positioning information to potential network-based radio-sensing services [2], the ability to understand the

communication status and predicted performance

- Value-adding auxiliary services such as computational device offloading [3] or AI as a service [4], services for distributing large amounts of data, or supporting digital twins.

These services have already been adopted in multiple ways, including CPaaS (communication platform as a service), which is used by large application developer communities to integrate different communication services to applications. These platforms offer APIs and integrated development environments that simplify the integration of communications capabilities (voice, messaging and video, for example) into applications and business processes.

Network platforms can exist at many levels. An operator’s network is a network platform that offers communication and other services. In many cases, there is also a need for global coordination, however, to provide ASPs with a single interface to a network service regardless of where the users of that application are located. This is particularly important for services that are independent of communications. Capabilities exposed from multiple network platforms can be aggregated into monetizable, globally available services for developers and enterprises. A platform like this acts as a network aggregator, as shown in Figure 1.

Network platforms are also related to other platforms in the mobile and information communications technology ecosystem, such as operating system platforms that expose underlying capabilities in phones; application platforms that focus on the distributing, governing and monetizing application distribution; and cloud compute and content delivery network platforms that enable flexible compute and content services.



Monetization and business support

Current expectations looking toward 2030 suggest a disconnect between the overall strong growth of the digital economy and the revenues of connectivity-centric telecom services. Creating a bridge between the connectivity and application business ecosystems will therefore be a key objective for 6G. Scaling 6G-empowered applications through distribution channels that have global reach (such as application platforms) would create synergy effects both for telecommunications and the application economy – covering application and application platforms, third-party ads and so on – spurring further growth in both sectors.

Creating a bridge between the connectivity and application business ecosystems is a key objective for 6G.

New services provided by the 6G platform must be designed to allow CSPs to increase and diversify their customer base. The new customers are expected to primarily consist of ASPs, verticals, application platform providers and other enterprises, and will enable CSPs to build API-based revenue streams. This will allow CSPs to create and capture new value connected to the digital, internet and application ecosystems as well as supporting a fast-paced digitalization through network buildout and densification.

In addition, for services beyond connectivity, CSPs can fully differentiate their offering based on the value delivered

and type of consumer. This will lead to opportunities to increase revenues and profitability, which in turn can support a positive spiral of investments in CSP service and infrastructure evolution.

Digitalization is not only a crucial factor for future economic growth, but also a key enabler for a more sustainable society. According to the climate exponential roadmap, digital services have the potential to enable a reduction in global emissions by 30-50 percent in the next decade [5]. A delay in network investment could endanger the transition toward a digital and more sustainable economy.

APIs are the means for the 6G platform to expose its capabilities to application developers, which is already established in 5G. By employing APIs, telecom infrastructure functionality can be integrated at application level, allowing for value-based charging models instead of the connectivity subscription model that is prevalent today. APIs for QoS, positioning, network compute offload and so on can add high value to the experienced quality of the applications, and it should therefore be possible to monetize them based on their value. This could be done by API access involving quality of experience (QoE) guarantees paid by the ASP, which in turn is monetized by application sales/subscriptions, third-party enterprise ads or by in-application purchases, for example. In addition, application distribution in mobile ecosystems today is dominated by application platforms tied to specific operating systems, notably Apple’s App Store and Google Play. There are indications that application developers would prefer to access network APIs through existing channels and integrate network API costs into application or in-application sales. Application platforms are already equipped to handle purchases that are consumed gradually, as well as subscriptions.

6G platform opportunities

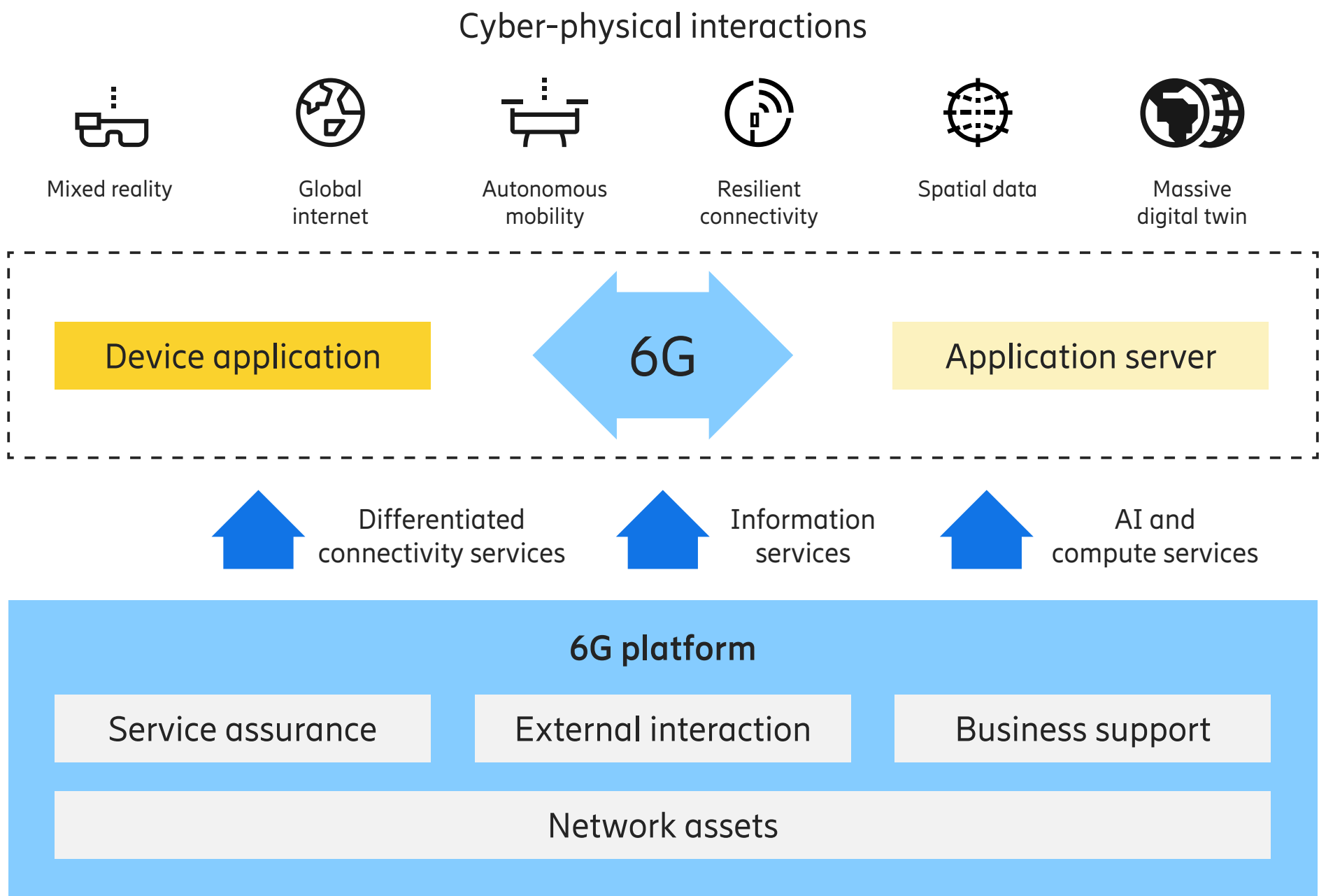


Figure 2: Advanced use cases expected to be supported by services from the 6G platform

CSPs are in a good position to offer these new services in their current local markets, but reaching a global market has historically proved challenging. Utilizing a network aggregator platform, however, CSPs can bootstrap into a global developer base.

Platform exposure through application programming interfaces

In the 6G time frame, the application developers, on both the server and client side, will have many more – and

more diverse – APIs available to choose from than they have today that expose both network and connectivity capabilities, as well as new services beyond connectivity. The 5G exposure solution must evolve to better support the 6G platform requirements, handle many more APIs, and provide new, customized APIs more easily and rapidly.

A key challenge for future network platforms is finding the right level, format and mechanism of exposing different platform services to involved ecosystem partners, including



private or enterprise developers, other platform providers and aggregators (network aggregators and application or cloud platforms), as well as device vendors. To other platforms and aggregators, network exposure through APIs evolving from current Service Capability Exposure Function/ Network Exposure Function (SCEF/NEF) approaches will be suitable (as is already ongoing in the GSMA's (GSM Association's) telco global API alliance CAMARA or in the 3GPP's (3rd Generation Partnership Project's) SA6). To reach application developers, however, it will be crucial to offer bundles of relevant APIs abstracted in libraries made accessible to developers as part of application-specific software development kits (SDKs) such as OpenXR, the device SDKs for Android and iOS, and so on.

The service innovation, creation and life-cycle management of APIs will be made simpler and faster with new supportive tools. Tools to assist API implementation, generation of documentation and SDKs, and customized example code used by the application developers. Other tools will support the service innovation process with technical, legal and business analyses. Technical analysis can support feature requirements, performance and reliability, for example. This may be used to upgrade and dimension the platform and to define Service Level Agreements (SLAs). The legal analysis can answer questions about conformance to the applicable legislation. The business analysis can estimate invocation cost and propose service level objectives, for example. Tools are also needed for customer onboarding and management of consumers and applications. One expected evolution is that service composition will be more flexible and simpler to use to create new services.

The platform will expose many standardized APIs, such as CAMARA APIs, to simplify aggregation of multiple 6G

platform providers. Today, REST is the dominant API stack (used by CAMARA APIs, for example) but we assume a more diverse situation with other stacks that have other characteristics, such as data query and manipulation language GraphQL and the cross-platform remote procedure call framework gRPC. Something more flexible and efficient for the exposure of data and data streams has yet to emerge.

Service setup and assurance

For services offered with performance and reliability commitments, there is also a need for mechanisms to assure fulfillment, both for the initial service setup and during the lifespan of the service. The CSP must be able to validate that the required resources are available, both for the new service and those already deployed. In case this cannot be done with sufficiently high confidence, service requirements must be adjusted to levels that can be fulfilled in the network. The process would vary depending on the customer. A large enterprise user could engage directly with the CSP to customize offerings and SLAs.

For common service types, an SLA portal may provide template SLAs that could be adjusted to specific enterprise needs. An ASP would define the requirements through the service API. On the other end, a private user who wants premium-performance gaming for an hour could use in-application purchases to request the add-on service realized by the application through the network API. Either way, the SLA would capture relevant requirements, such as a minimum throughput and a maximum latency, or a consumer QoE. For add-on services, it can also include positioning accuracy, sensing resolution, or processing capacity for compute offload.

Tools such as slicing, time-critical communication and

6G platform opportunities

QoS frameworks [6] will be used to configure the needed traffic flows in the network. On the device side, user equipment (UE) route selection policy (URSP) can be used to identify the desired application and map its traffic to the configured traffic flows. In addition, resources for beyond-communication services will be allocated. Once setup is complete, the dynamic network environment needs to be managed. Performance will be impacted both by the overall traffic situation with congestion points occurring at different places at different times. An individual user may be in a bad radio location, or a cell may need to serve more users than originally anticipated. To manage this, the 6G platform automatically instantiates assurance processes at different levels to adjust network resources to achieve the desired performance.

The introduction of intents provides a structured way of capturing both service and network requirements, and their relative importance.

The ability to observe relevant key performance indicators is instrumental. Metrics are gathered that directly or indirectly reflect the performance objectives. For instance, latency and throughput could be used to estimate the QoE. It could also be the case that an offloaded rendering process is hitting the

capacity limit in the compute infrastructure. If we identify that QoE is degrading due to network congestion in the radio access network, the assurance process in the radio base station could adjust resource allocation or scheduling parameters to improve the needed performance metric. In doing so, we need to balance against requirements from other services and objectives on network operation at large, such as energy consumption.

The introduction of intents [7] provides a structured way of capturing both service and network requirements, and their relative importance. This makes it possible to effectively prioritize and ensure performance is optimized even in situations of congestion or failures. In most cases, intents would not be directly exposed to consumers. Instead, they would use an API or service portal, where the needed information would be captured and translated into intents that can be complemented with operator-specific objectives reflecting the commercial offering, the network deployment and operational priorities.

Example use case: exposed 6G platform services for mixed reality

To illustrate how the different 6G platform components will work together, we will use the example of an immersive MR application developed on an augmented reality (AR)/ MR application platform. The MR application is experienced using a lightweight, mobile head-mounted display (HMD) supported by cameras and a variety of sensors. Through the HMD, digital representations of communication partners, virtually generated personas or digital objects are integrated into the user's view.

While such an immersive experience requires the complex interworking of many services, the user will judge the quality

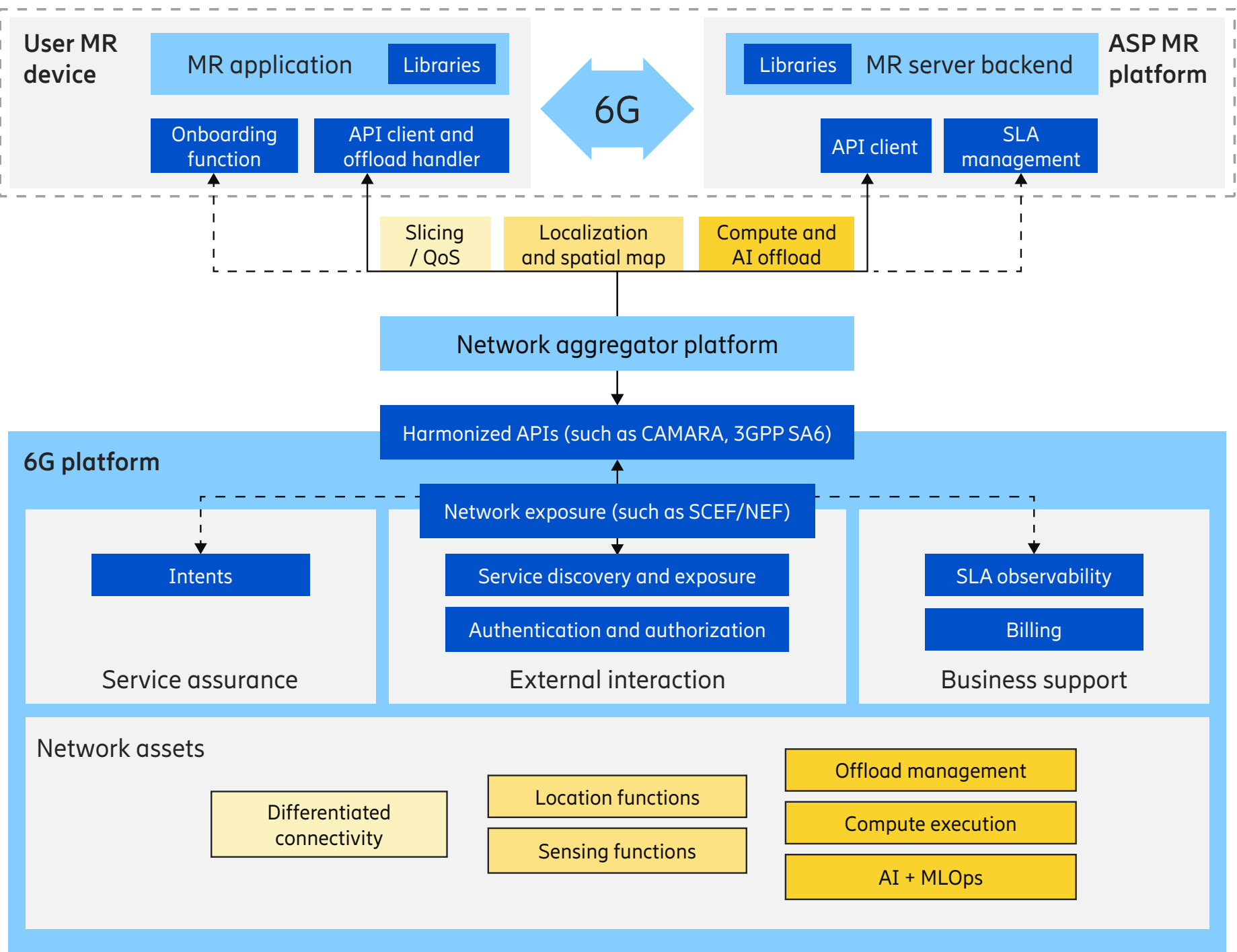


Figure 3: An MR application supported by services from the 6G platform

of the overall experience mostly by the steady, seamless integration and smooth interaction of the virtual objects with the real world and the fidelity of the graphics, as well as the convenience of wearing the HMD in a mobile scenario. Good QoE will reflect favorably on the ASP's offering. Both the ASP and the consumer have an incentive to use the best communication and other services to achieve the necessary QoE.

Figure 3 illustrates how an immersive MR application benefits from an appropriate bundle of 6G platform services to facilitate smooth application performance. The ASP initiates tailored connectivity with uplink/downlink capacity and latency that matches the requirements of the MR application [8]. To realize this in the networks, URSP is used to route the MR application traffic to a network slice that is dedicated for this purpose (as already demonstrated successfully in 5G [9]).

MR applications are heavily dependent on positioning and orientation information to accurately map digital content in relation to the real world. In our example, the MR application will be able to take advantage of the information services offered by the 6G platform, specifically (active) positioning and (passive) environmental sensing by the network. This information will enrich existing sensor data (for instance from gyroscopes, video, GNSS (Global Navigation Satellite System)) and may be consumed in different grades of processing (raw data, a basic digital map around the user's location or a spatial mapping of objects within the digital map, for example) depending on the preferences of the AR/MR application. For instance, rough positioning by GNSS and a network positioning result could be complemented with a spatial map of the local environment, allowing the application to do quick and exact localization based on visual means.

The MR application can further benefit from a network-integrated compute service that offers the possibility to seamlessly offload and place processing-intensive tasks within the compute continuum stretching from the device via edge compute to the cloud. In the case of a typical MR application, this applies to tasks such as map optimization or object detection [10]. Based on the compute/heat/battery conditions of the HMD, radio conditions and the available remote compute capacities, an offloading service would determine when and where it is beneficial to dynamically offload some of these tasks for the benefit of optimized application and device experience. The optimization criteria would be based on policies set by the user, the ASP, the CSP and potentially even the device vendor. By adding a small offload handler to the device (either as an application, operating-system function or part of the network stack) and reusing network authentication and authorization

mechanisms, the resulting dynamic computational offload service would be consumed alongside the connectivity-centric APIs of the network platform to ensure a richer experience for users.

Finally, the MR application uses the exposed service APIs to request the currently desired service characteristics and to learn what the available service is from the network in the current situation. The network then captures the agreed service in the form of intents, initiates the service, runs the needed assurance process, and so on. In short, the network platform strives to ensure a seamless application and device experience, and the application has the necessary information and support to tune its behavior for the best user experience.

Conclusion

The 6G platform will expand the scope of networks beyond communication services to support the services needed by applications, which are at the center of the digital transformation. Application service providers will have easy access to network capabilities and a wide range of service application programming interfaces. Consumers will benefit from enhanced quality of experience, while communication service providers gain the opportunity to expand their role in the ecosystem beyond bit-pipes, with new possibilities for monetization higher up in the value chain.



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